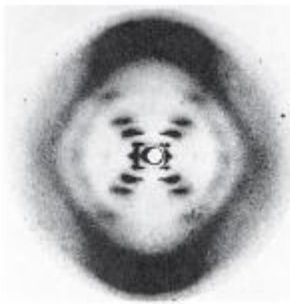
UNIVERSIDAD PERUANA
CAYETANO HEREDIA

Course Molecular Biology for Engineers

DNA Elements of the Human Genome

Deoxyribonucleic acid structure



Rosalind Franklin / Photo 51

James Watson/ Francis Crick

No. 4356 April 25, 1953 NATURE
737

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three inter-twined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion. this structure is unsatisfactory for two reasons:
(1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions. namely, that each chain consists of phosphate diester groups joining ß-D-deoxy-ribofuranose residues with 3',5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow righthanded helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions.

This figure is purely diagrammatic. The two ribbons symbolize the two phosphate—sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain, does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain, is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in time following, communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

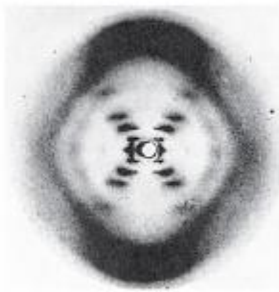
We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J.D.W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

J.D. WATSON
F.H. C. CRICK

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

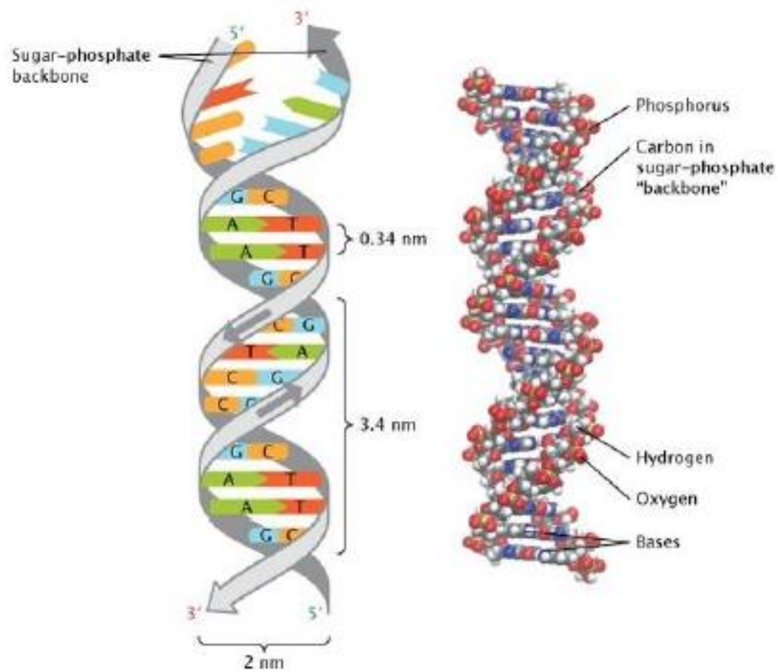
[1] Pauling, L., and Corey, R. B. nature, 171, 346 (1953); Proc, U.S. Nat Acad. Sci., 39, 84 (1953).
[2] Furberg, S., Acta Chem. Scand., 6, 634 (1952).
[3] Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., Biochim, et Biophys. Acta, 9 402 (1952).
[4] Wyatt, G.R. J. Gen. Physiol., 36 201 (1952).
[5] Astbury, W.T., Symp. Soc. Exp. Biol. 1, Nucleic Acid, 66 (Camb. Univ. Press, 1947)
[6] Wilkins, M. H. F. and Randall, J. T. Biochim. et. Biophys. Acta, 10, 102 (1953).

Deoxyribonucleic acid structure

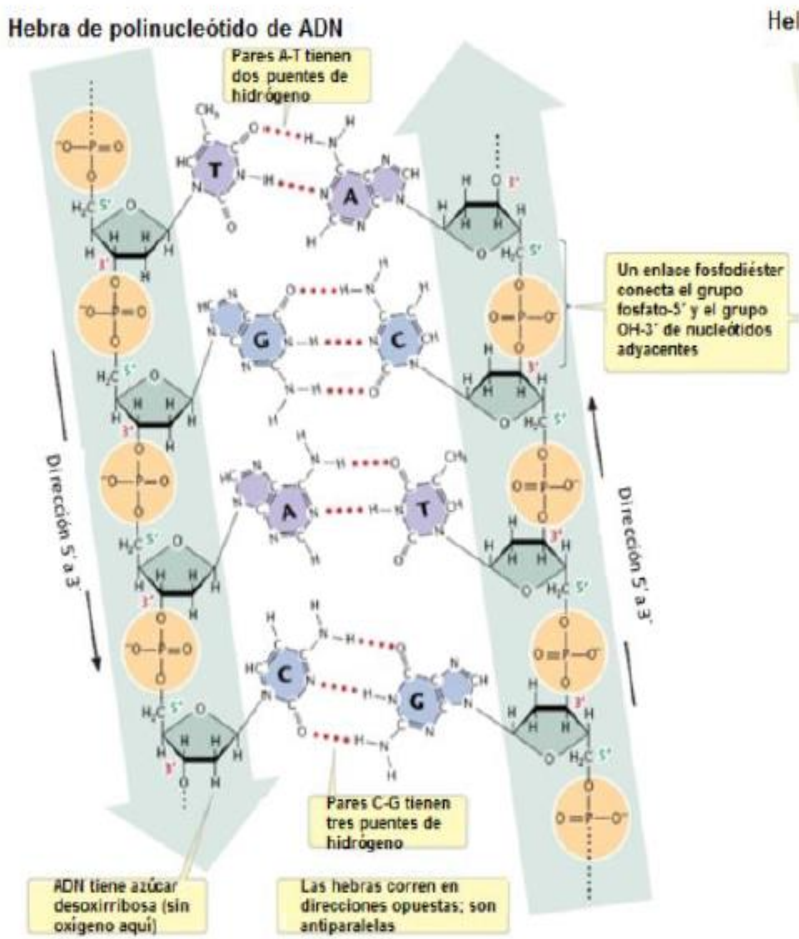
Rosalind Franklin / Photo 51

James Watson/ Francis Crick



Pray, L. (2008) Discovery of DNA structure and function: Watson and Crick. *Nature Education* 1(1):100
https://www.nature.com/scitable/topicpage/discovery-of-dna-structure-and-function-watson-397/

Deoxyribonucleic acid structure



Double stranded DNA has antiparallel strands. Why?

5'→3'

3'→5'

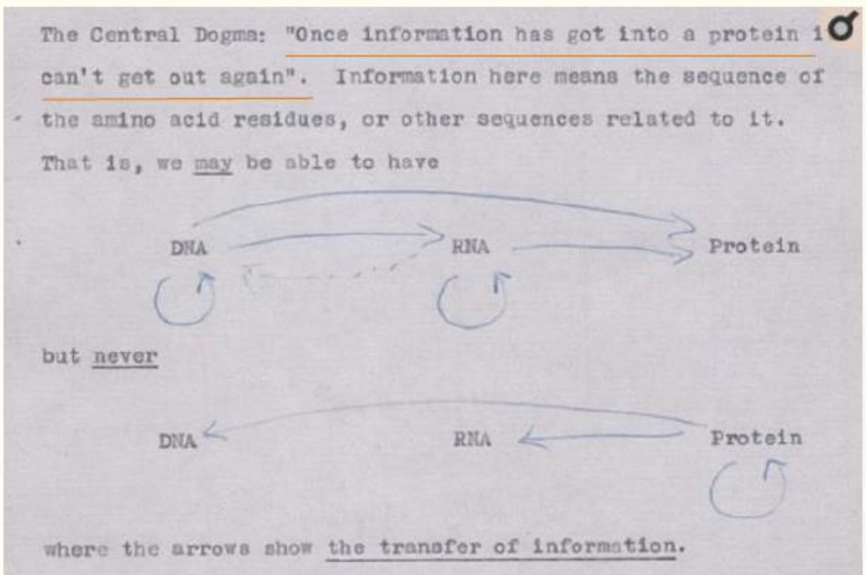
The bases are complementary to each other.

Thymine - Adenine

Guanine - Cytosine

	$\Delta G'$
A-T	-12.4 kcal/mol
G-C	-25.4 kcal/mol

Central dogma of Molecular Biology


The Central Dogma: "Once information has got into a protein i can't get out again". Information here means the sequence of the amino acid residues, or other sequences related to it.

That is, we may be able to have

DNA RNA Protein

but never

DNA RNA Protein

where the arrows show the transfer of information.

Crick's first outline of the central dogma, from an unpublished note made in 1956.

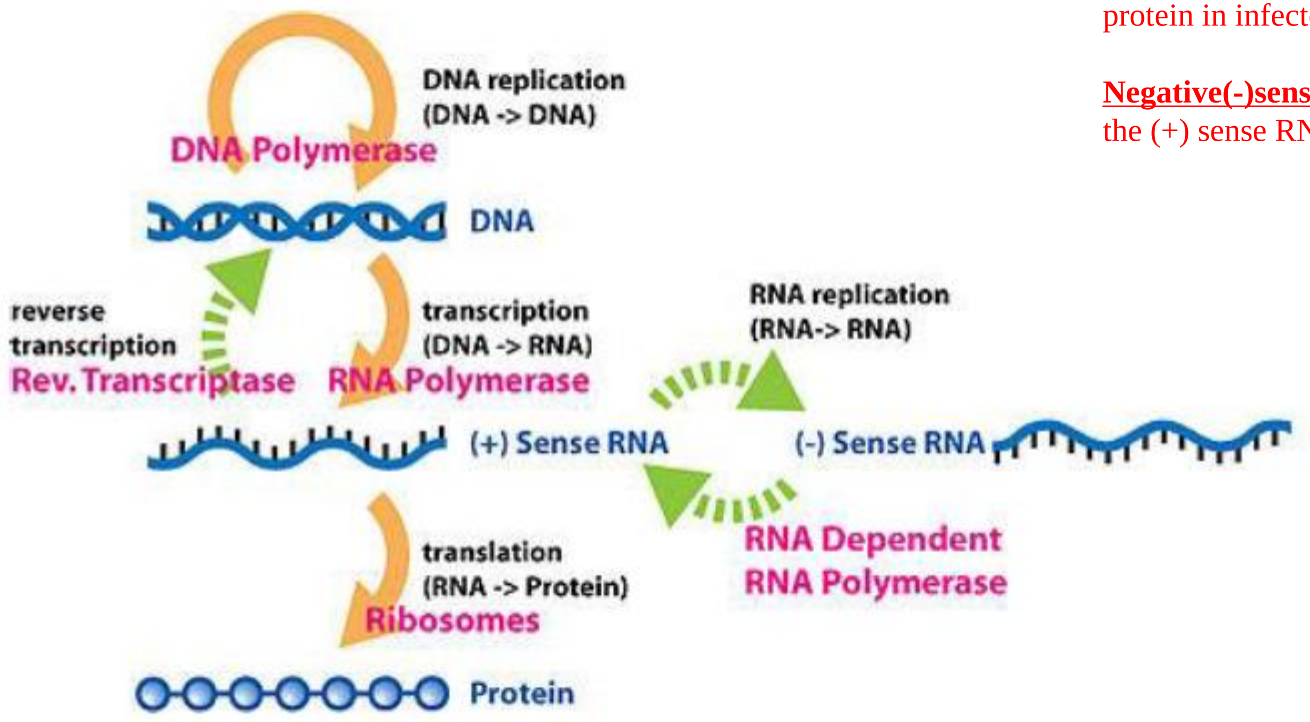
The Central Dogma

This states that once 'information' has passed into protein *it cannot get out again*. In more detail, the transfer of information from nucleic acid to nucleic acid, or from nucleic acid to protein may be possible, but transfer from protein to protein, or from protein to nucleic acid is impossible. Information means here the *precise* determination of sequence, either of bases in the nucleic acid or of amino acid residues in the protein.

Crick FH (1958). On protein synthesis. Symp. Soc. Exp. Biol. 12: 138–63

"I shall…argue that the main function of the genetic material is to control (not necessarily directly) the synthesis of proteins. There is a little direct evidence to support this, but to my mind the psychological drive behind this hypothesis is at the moment independent of such evidence."

GENETIC INFORMATION FLOW (CENTRAL DOGMA)



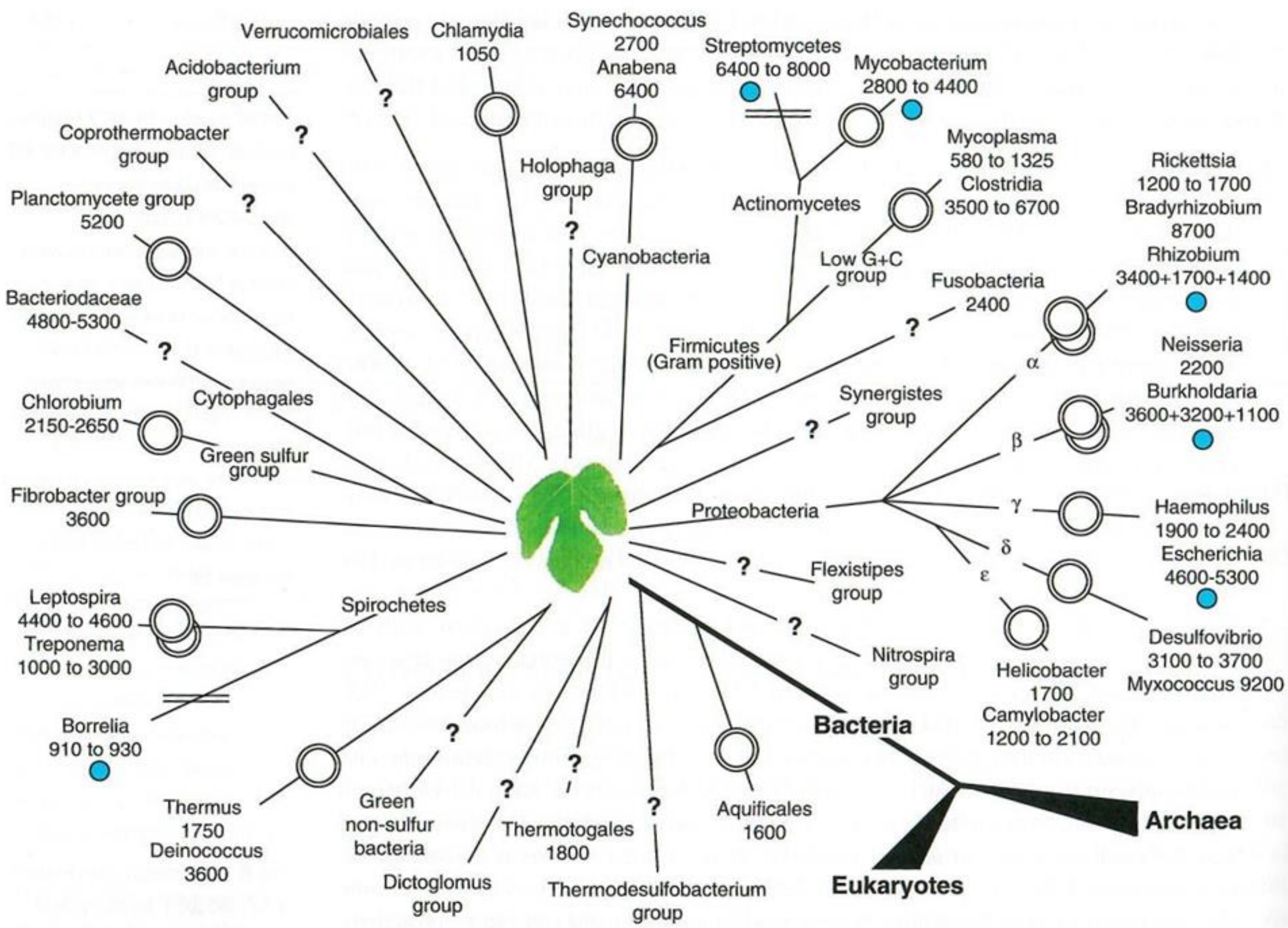
<u>Positive (+) sense RNA</u>: RNA that can directly be translated into proteins. Positive-sense RNA virus like Sars-Cov-2 have a (+) sense RNA that is translated into protein in infected cells.

<u>Negative(-)sense RNA</u>: RNA that has to be synthesized the (+) sense RNA to get translated into proteins.

Figure taken from: https://en.wikipedia.org/wiki/Central_dogma_of_molecular_biology

Bacteria genomes

- Bacteria genomes are generally small circular structures
- Size: Ranging from 0.6 to 15 Mb in length
- Encodes 600–6,000 genes



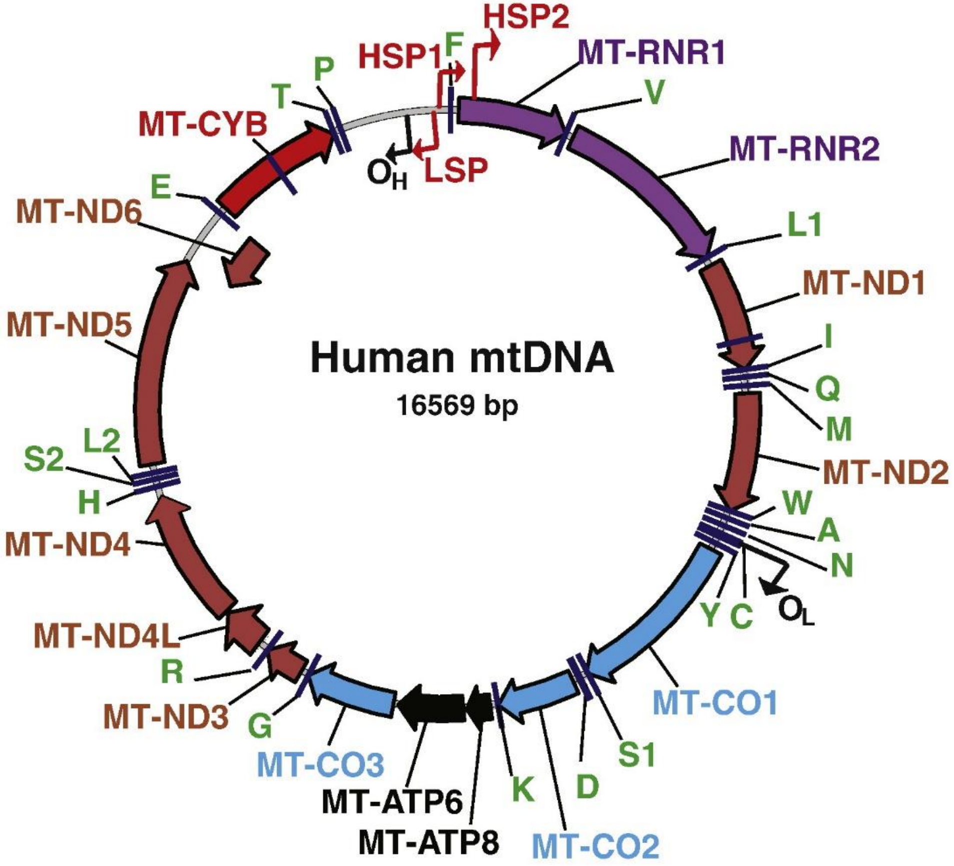
Human genome

- 23 chromosome pairs in the cell nuclei. 22 somatic chromosome pairs, 1 sex pair chromosomes (X, Y)
- Size: 3,000, 0000,0000 bp (3 Gbp, 3 x 10^9 bp)
- 25,000 genes

Nurk, S., Koren, S., Rhie, A., Rautiainen, M., Bzikadze, A. V., Mikheenko, A., ... & Phillippy, A. M. (2022). The complete sequence of a human genome. Science, 376(6588), 44-53.

Human Mitochondrial genome

Circular genome. Size: 16.5 kb
37 genes: 13 polypeptides and 24 RNA components of the mitochondrial translational apparatus (2 rRNAs and 22 tRNAs).



Shokolenko, I. N., & Alexeyev, M. F. (2015). Mitochondrial DNA: A disposable genome?. *Biochimica et Biophysica Acta (BBA)-Molecular Basis of Disease*, *1852*(9), 1805-1809.

Genome Size and gene number

	Prokaryote		Eukaryote			
	E.coli	***M. tuberculosis***	***Yeast***	***P. falciparum***	***S. mansoni***	***H. sapiens***
Genome Size (Mb)	5	4	13	23	363	3,300
GC (%)	51	66	38	19	35	41
Gene number	4,300	3,924	5,770	5,628	11,809	25,000
Gene density (kb/gene)	1	1	2	4	29	27
Coding sequence (% genome)	88	91	71	53	5	1
Intron number	0	0	272	7,400	6,000	55,000
Repeat sequences (%)	<1	<1	2	<1	45	46

EUKARYOTIC CHROMOSOMES

Linear DNA sequences

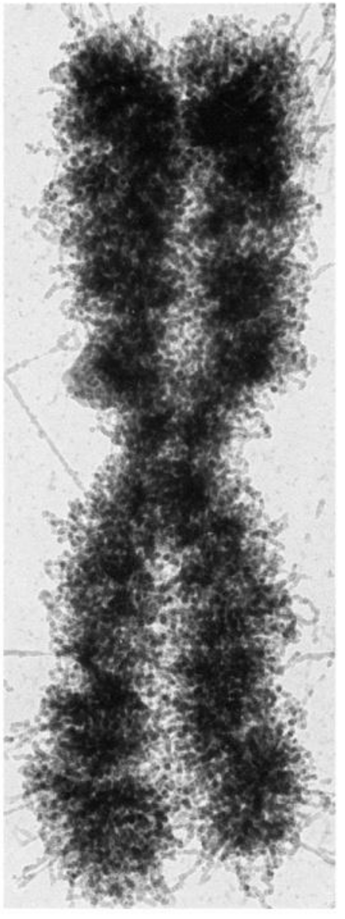
Multicellular organisms (human, mouse):

- Condense in metaphase during cell division
- Karyotype FISH

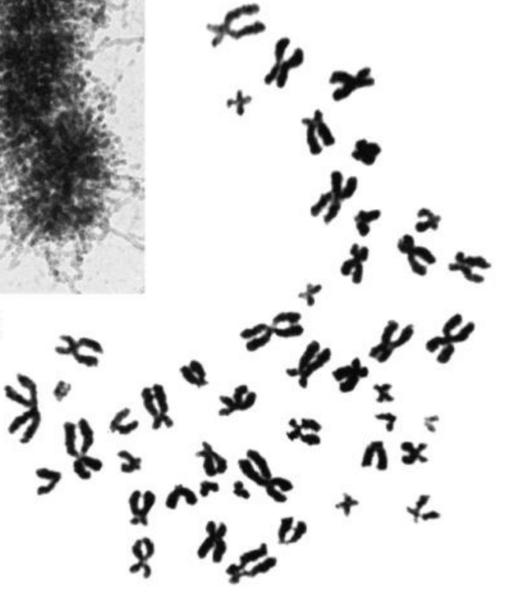
Protozoa, single-celled eukaryotes (yeast, plasmodium, trypanosome, amoeba:

- No evidence of condensation during cell division

- Karyotype by Pulse Field Gradient Electrophoresis (PFGE)



(a)



(b)

Organization of DNA Elements in Eukaryotic cells

Linear chromosomes

Centromere:

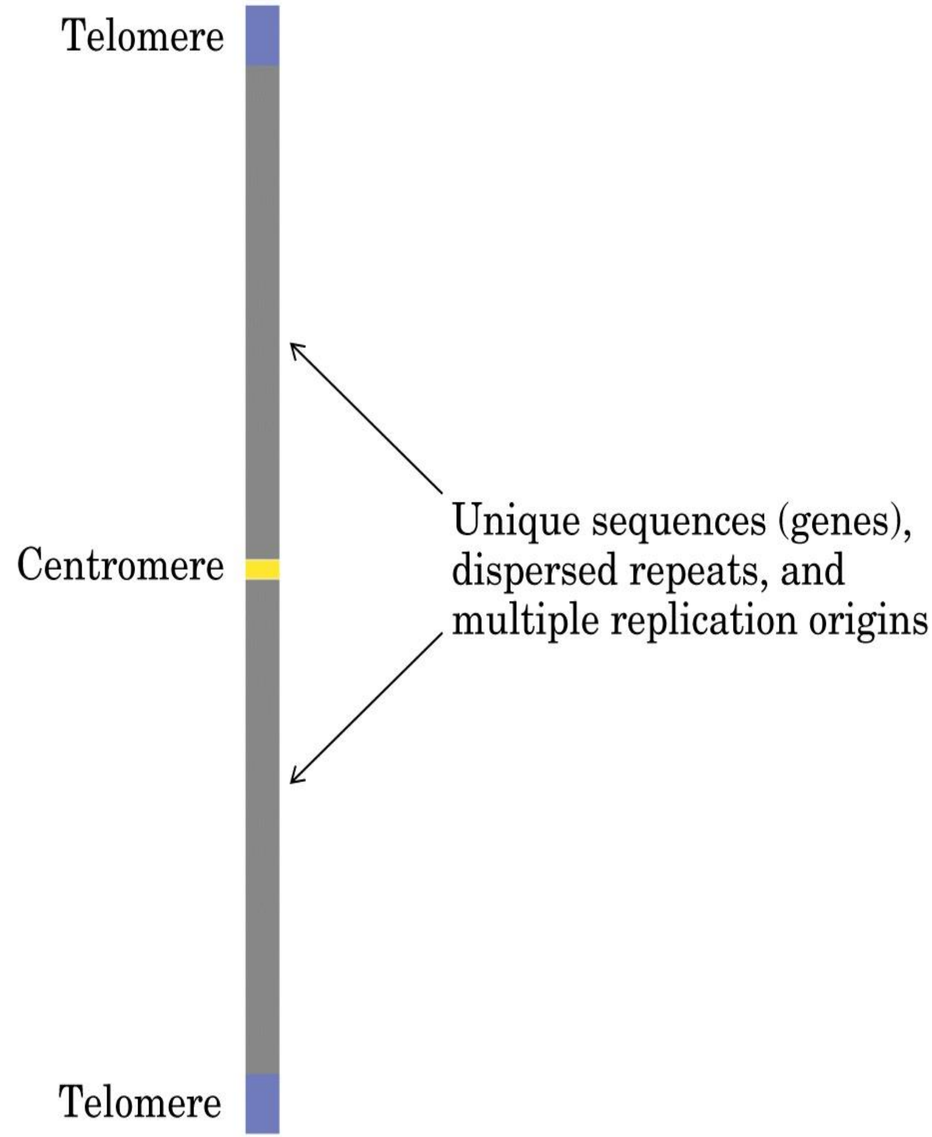
Function:

- Chromosome segregation during cell división.

Telomere:

Function:

- Protection of ends of chromosomes from degradation
- Stability
- Replication, avoids shortening of chromosomes during replication



Telomeres

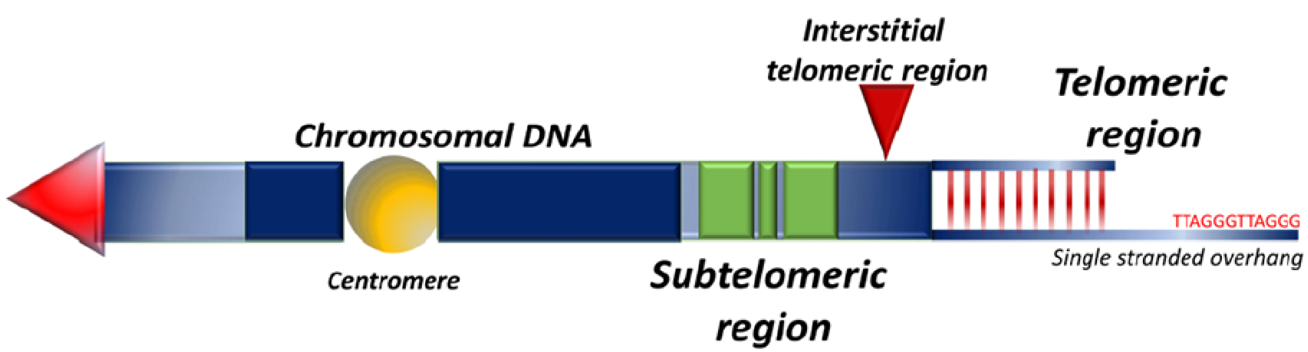
- Telomeres are formed by tandem repetitive hexameric sequence located at the ends of all human chromosomes.

- Human telomeres typically range between 10 to 15 kb of the highly conserved tandem nucleotide repeats is 5′- TTAGGG-3′ .

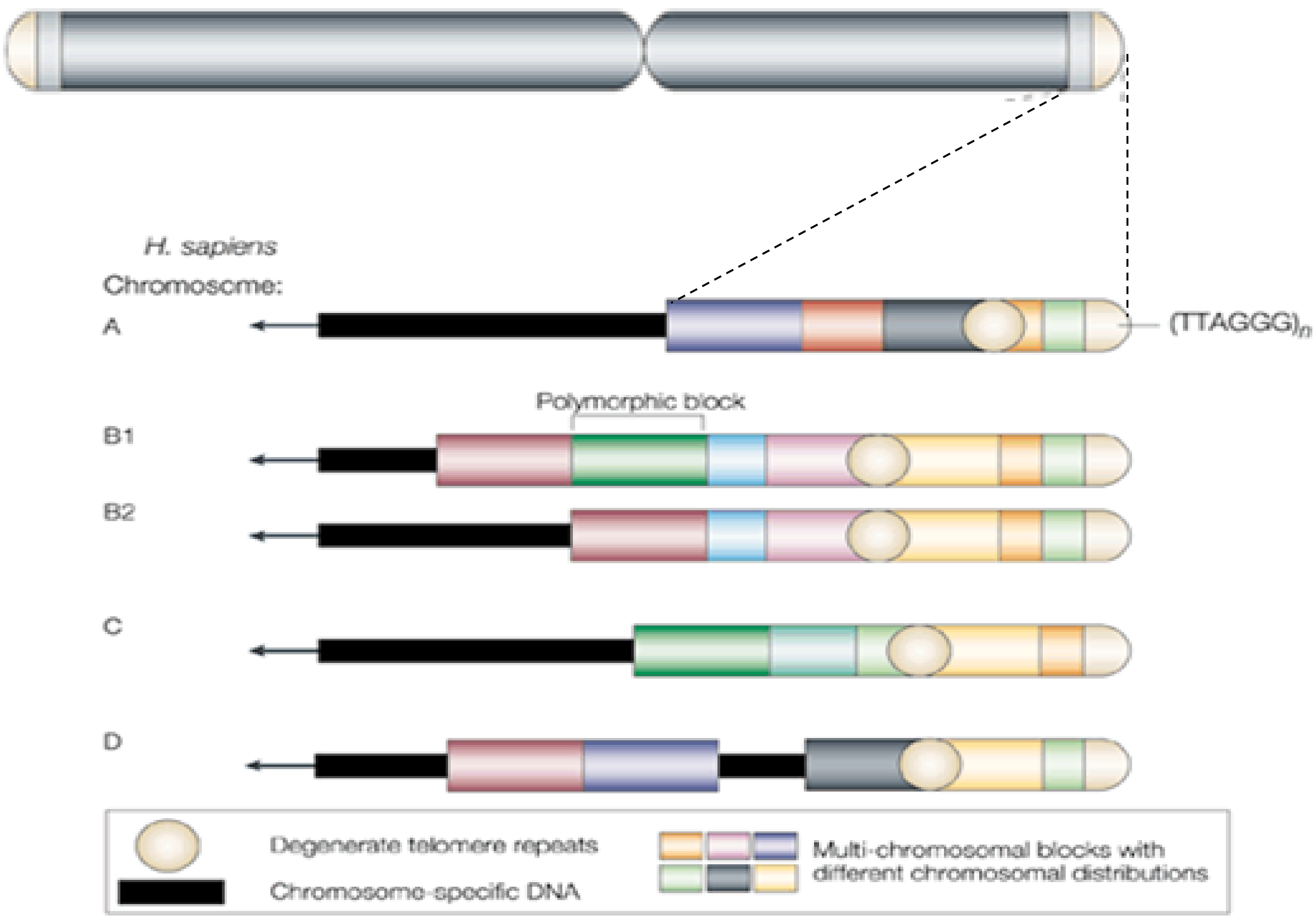
- The telomere form a complex structure with shelterin proteins for the protection at chromosomal ends to maintain genomic integrity.

- Next to telomere is the subtelomere formed for more complex repetitive sequences.

- Telomeres are replicated during cell division by the specialized ribozyme Telomerase.



Human Telomeres



Nature Reviews | Genetics

Human Telomere Replication

- Telomere shortens as cells divide because DNA polymerases cannot fully replicate linear chromosomes.
- Telomeres are replicated during cell division by the specialized ribozyme Telomerase
- Telomere length dynamics is associated with aging and senescence. Critically short telomeres initiate cellular senescence, arresting cell division.
- Telomere shortening is counteracted by telomerase, a reverse transcriptase that uses an RNA template telomerase RNA component (TERC), to synthesize new telomeric repeats.
- TERC is a 451-nucleotide long non-coding RNA that acts as a template for telomere synthesis.

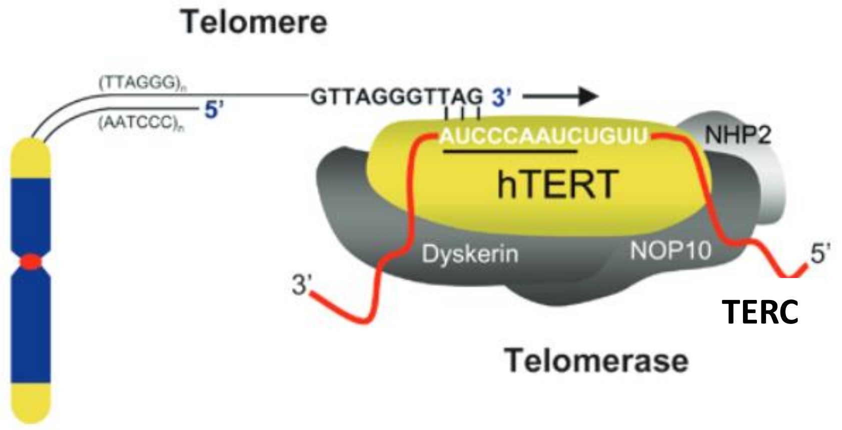


Fig. 16.2 *Structure and function of telomerase.* The human telomerase complex consists of a catalytic subunit (hTERT), an RNA component (hTR), dyskerin, NOP2, NHP2, and additional associated proteins. Here, the de novo synthesis of telomeric repeats by telomerase is illustrated. The template region of hTR is *underlined*. Four additional nucleotides complementary to the 3'-end of imetelstat are also shown

Röth et al. (2009). Imetelstat (GRN163L)-telomerase-based cancer therapy. *Small Molecules in Oncology*, 221-234.

Telomere length and aging

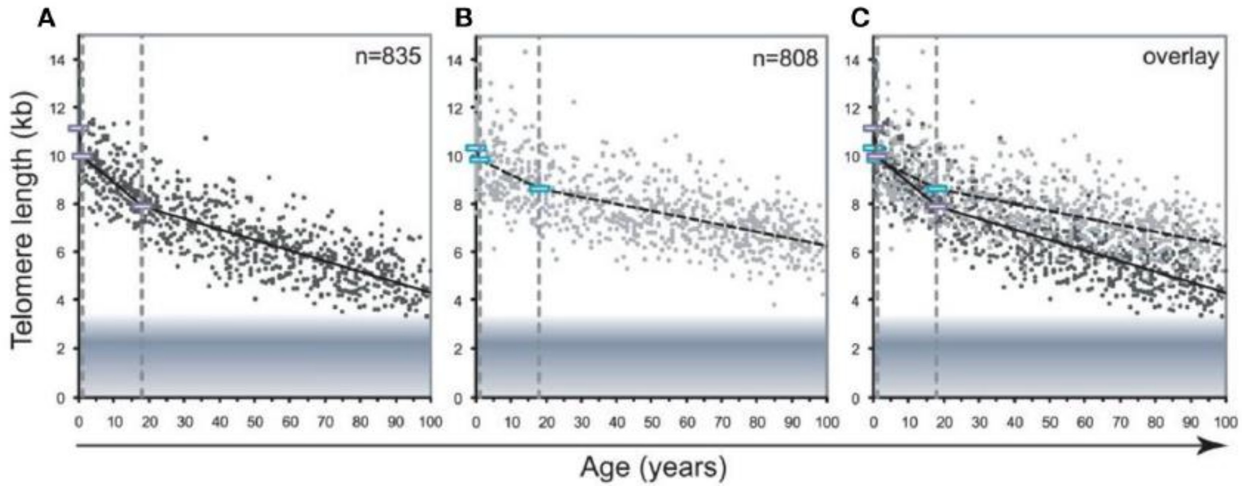


FIGURE 9 | Decline in telomere length with age differs between lymphocytes and granulocytes. The median telomere length in nucleated blood cells from 835 healthy individuals ranging from birth (umbilical cord blood) to 102 years of age were measured by flow FISH. The results were used to calculate the telomere attrition over time using linear regression in three age segments. **(A)** Median telomere length in lymphocytes (black dots). **(B)** Median telomere length in granulocytes (gray dots). Breakpoints in the piece-wise linear regression lines are marked by rectangles and the three age groups are marked by dotted vertical gray lines at 1 and 18 years. On average 8 individuals were tested per age-year. **(C)** At any given age, a wide range of telomere length was observed and the decline in telomere length with age in lymphocytes was more pronounced than in granulocytes. The shaded area represents the estimated length of subtelomeric DNA. Note that in older individuals, on average only 1–2 kb of telomere repeats were present in lymphocytes. The figure and its legend are reproduced from the article by Aubert et al. (2012a) distributed under the terms of the Creative Commons Attribution License. Copyright: © 2012 Aubert et al.

Vaiserman, A., & Krasnienkov, D. (2021). Telomere length as a marker of biological age: state-of-the-art, open issues, and future perspectives. *Frontiers in genetics*, *11*, 630186.

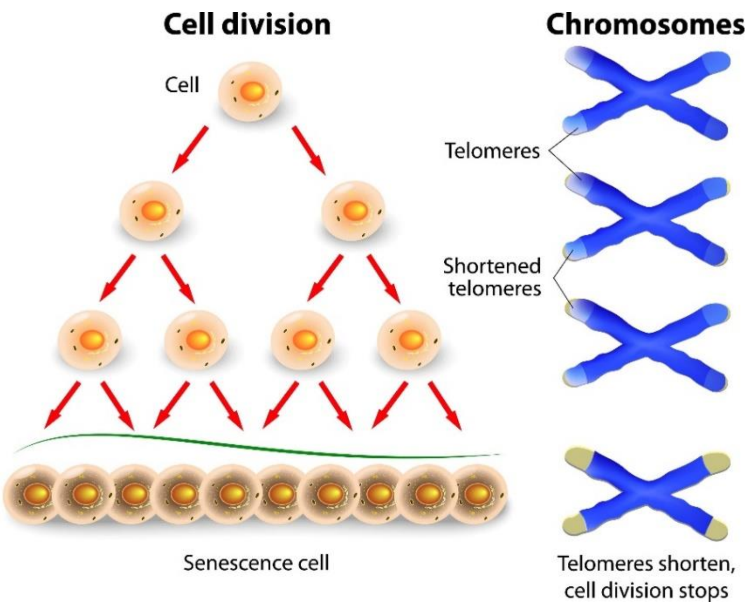
Calado, R. T., & Young, N. S. (2009). Telomere diseases. *New England Journal of Medicine*, *361*(24), 2353-2365.

Hayflick limit, telomere shortening and senescence

Hayflick limit refers to the number of times a normal cell population can divide before entering the senescence phase, typically ranging between 40 and 60 divisions in human fetal cells.

This phenomenon is associated with telomere shortening, which ultimately restricts further cell division and correlates with aging.

Once the cell reaches the Haflick limit stops dividing but stays alive.

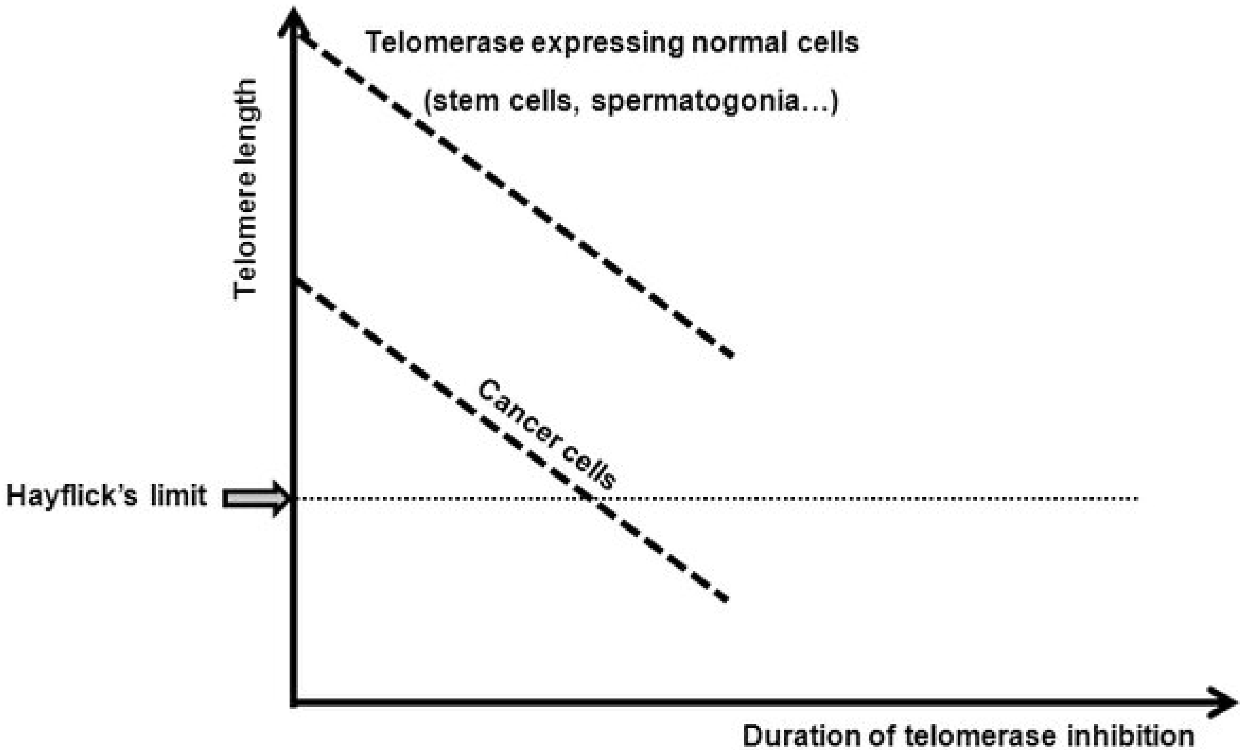


Hayflick limit is reached when telomeres shorten to a critical threshold, typically around **5 to 7 kilobase pairs (kb)** length.

Why telomeres are relevant to Biomedical Engineering (BME)?

Feature	Normal Somatic Cell	Cancer/Stem Cell
Telomere Status	Shortens with every division	Maintained by telomerase
Replication Potential	Limited (approx. 50–70 divisions)	Theoretically infinite
BME Focus	Preventing premature aging	Inhibiting growth / Detection

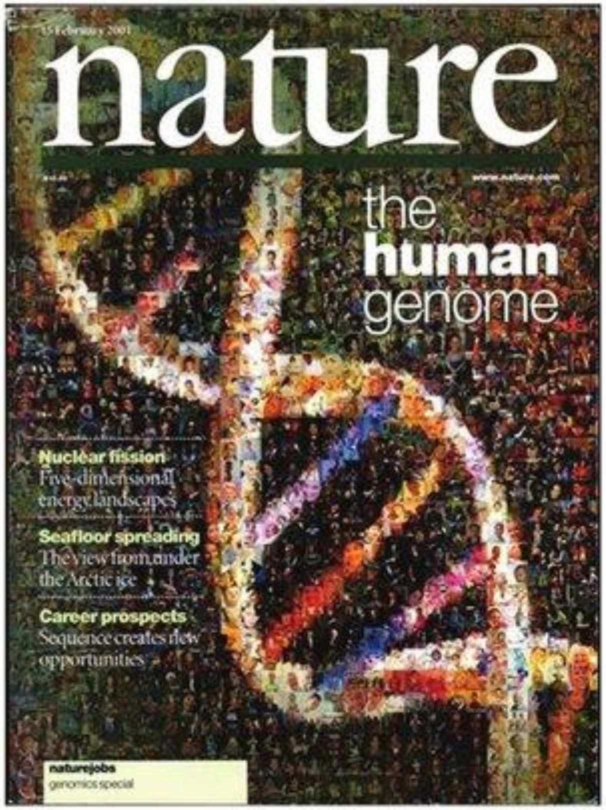
Inhibition of telomerase in cancer treatment

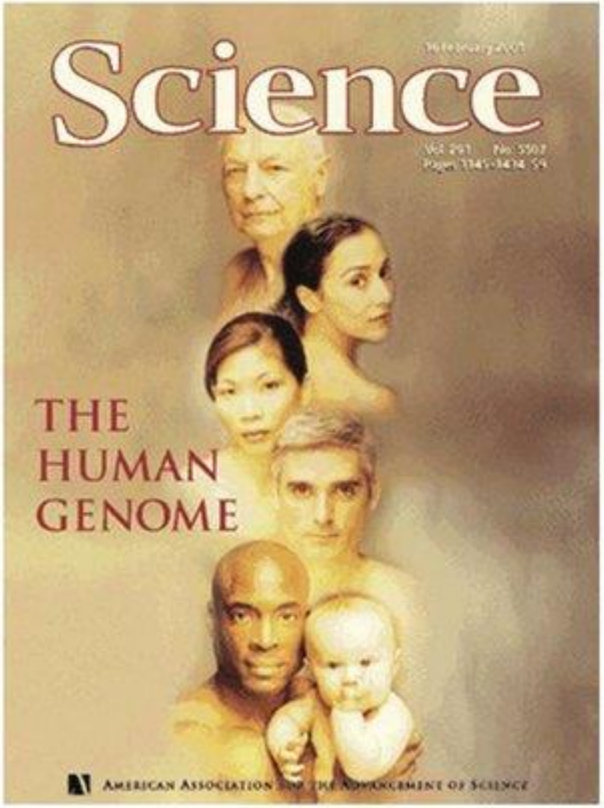


- Upon telomerase inhibition, telomeres in cancer cells will reach the Hayflick's limit faster than the normal cells, thus enabling a reasonable therapeutic window for the use of telomerase inhibitors.
- Imetelstat is a 13-mer oligonucleotide complementary to the TERC template region, which competitively inhibits telomerase activity, suppressing cancer cell viability in vitro and tumour growth in mouse xenograft models

Uziel, O., & Lahav, M. (2014). Conventional anticancer therapeutics and telomere maintenance mechanisms. Current Pharmaceutical Design, 20(41), 6452-6465.

FIRST COMPLETE SEQUENCE OF THE HUMAN GENOME

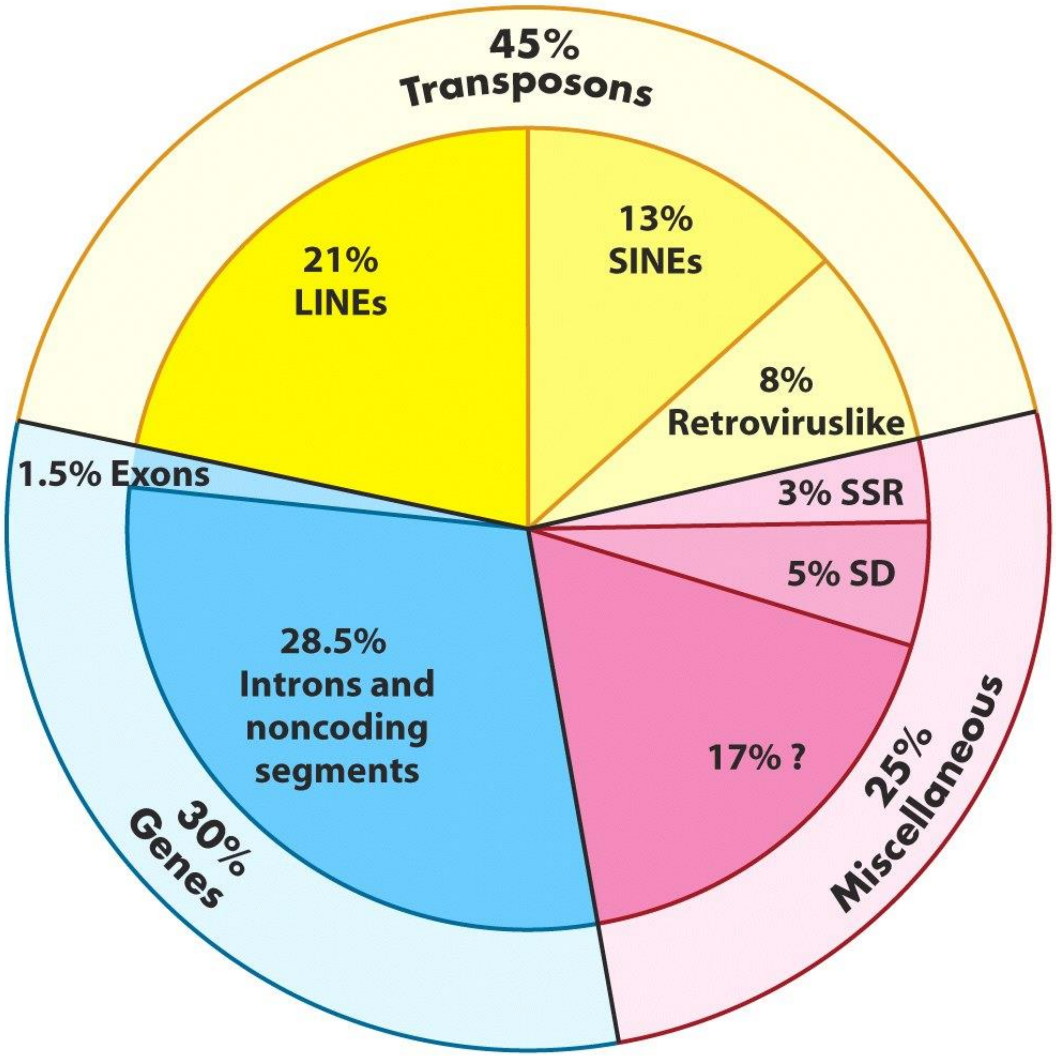




June 2000

DNA ELEMENTS OF THE HUMAN GENOME

- 3,000, 0000,0000 bp (3 Gbp, 3 x 10^9 bp)



June 2000

RESEARCH ARTICLE

HUMAN GENOMICS

The complete sequence of a human genome

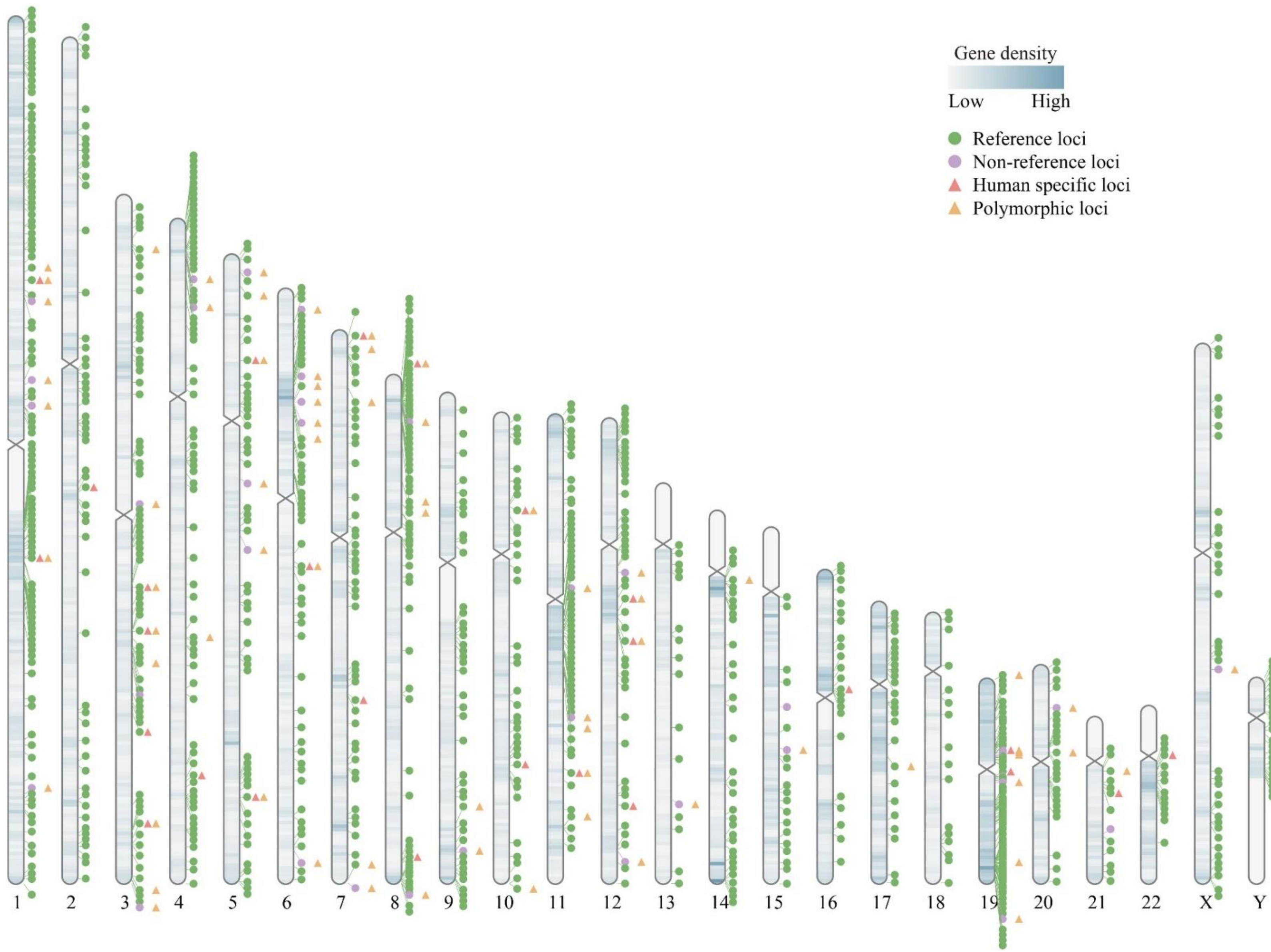
Sergey Nurk[1]†, Sergey Koren[1]†, Arang Rhie[1]†, Mikko Rautiainen[1]†, Andrey V. Bzikadze[2], Alla Mikheenko[3], Mitchell R. Vollger[4], Nicolas Altemose[5], Lev Uralsky[6,7], Ariel Gershman[8], Sergey Aganezov[9]‡, Savannah J. Hoyt[10], Mark Diekhans[11], Glennis A. Logsdon[4], Michael Alonge[9], Stylianos E. Antonarakis[12], Matthew Borchers[13], Gerard G. Bouffard[14], Shelise Y. Brooks[14], Gina V. Caldas[15], Nae-Chyun Chen[9], Haoyu Cheng[16,17], Chen-Shan Chin[18], William Chow[19], Leonardo G. de Lima[13], Philip C. Dishuck[4], Richard Durbin[19,20], Tatiana Dvorkina[3], Ian T. Fiddes[21], Giulio Formenti[22,23], Robert S. Fulton[24], Arkarachai Fungtammasan[18], Erik Garrison[11,25], Patrick G. S. Grady[10], Tina A. Graves-Lindsay[26], Ira M. Hall[27], Nancy F. Hansen[28], Gabrielle A. Hartley[10], Marina Haukness[11], Kerstin Howe[19], Michael W. Hunkapiller[29], Chirag Jain[1,30], Miten Jain[11], Erich D. Jarvis[22,23], Peter Kerpedjiev[31], Melanie Kirsche[9], Mikhail Kolmogorov[32], Jonas Korlach[29], Milinn Kremitzki[26], Heng Li[16,17], Valerie V. Maduro[33], Tobias Marschall[34], Ann M. McCartney[1], Jennifer McDaniel[35], Danny E. Miller[4,36], James C. Mullikin[14,28], Eugene W. Myers[37], Nathan D. Olson[35], Benedict Paten[11], Paul Peluso[29], Pavel A. Pevzner[32], David Porubsky[4], Tamara Potapova[13], Evgeny I. Rogaev[6,7,38,39], Jeffrey A. Rosenfeld[40], Steven L. Salzberg[9,41], Valerie A. Schneider[42], Fritz J. Sedlazeck[43], Kishwar Shafin[11], Colin J. Shew[44], Alaina Shumate[41], Ying Sims[19], Arian F. A. Smit[45], Daniela C. Soto[44], Ivan Sović[29,46], Jessica M. Storer[45], Aaron Streets[5,47], Beth A. Sullivan[48], Françoise Thibaud-Nissen[42], James Torrance[19], Justin Wagner[35], Brian P. Walenz[1], Aaron Wenger[29], Jonathan M. D. Wood[19], Chunlin Xiao[42], Stephanie M. Yan[49], Alice C. Young[14], Samantha Zarate[9], Urvashi Surti[50], Rajiv C. McCoy[49], Megan Y. Dennis[44], Ivan A. Alexandrov[3,7,51], Jennifer L. Gerton[13,52], Rachel J. O'Neill[10], Winston Timp[8,41], Justin M. Zook[35], Michael C. Schatz[9,49], Evan E. Eichler[4,53]*, Karen H. Miga[11,54]*, Adam M. Phillippy[1]*

Since its initial release in 2000, the human reference genome has covered only the euchromatic fraction of the genome, leaving important heterochromatic regions unfinished. Addressing the remaining 8% of the genome, the Telomere-to-Telomere (T2T) Consortium presents a complete 3.055 billion–base pair sequence of a human genome, T2T-CHM13, that includes gapless assemblies for all chromosomes except Y, corrects errors in the prior references, and introduces nearly 200 million base pairs of sequence containing 1956 gene predictions, 99 of which are predicted to be protein coding. The completed regions include all centromeric satellite arrays, recent segmental duplications, and the short arms of all five acrocentric chromosomes, unlocking these complex regions of the genome to variational and functional studies.

Nurk, S., Koren, S., Rhie, A., Rautiainen, M., Bzikadze, A. V., Mikheenko, A., ... & Phillippy, A. M. (2022). The complete sequence of a human genome. *Science*, *376*(6588), 44-53.

31 March 2022

GENE DISTRIBUTION IN THE HUMAN GENOME



DNA ELEMENT	GENOME (T2T-CHM14)	GENOME SEQ (%)
Assembled bases (Gbp)	3.05	**100%**
Annotation		
Number of genes	63,494.00	
Protein coding (Mbp)	19,969.00	0.65%
Number of transcripts	233,615.00	
Protein coding (Mbp)	86,245.00	2.83%
Total protein coding		**3.48%**
Duplications		
Segmental duplication bases (Mbp)	201.93	6.62%
Number of segmental duplications	41,528.00	
Total Duplications		**6.62%**
Repeat Masker		
Repeat bases (Mbp)	1,647.81	53.94%
Mobile Elements		
Long interspersed nuclear elements (LINE)	631.64	20.7%
Short interspersed nuclear elements (SINE)	390.27	12.8%
Long terminal repeats (LTR)	269.91	8.8%
Retroposon	4.65	0.2%
DNA transposon (relics)	109.35	3.6%
Total DNA Mobile elements (Mbp)	1,296.47	**46.09%**
Tandem Repetitive Elements		
Satellite	150.42	4.93%
Simple repeat (SSR)	77.69	2.55%
		7.48%

Organization of DNA Elements in Eukaryotic cells

Gene

Eukaryotic genes have segments of coding sequences (exons) interrupted by noncoding sequences (introns).

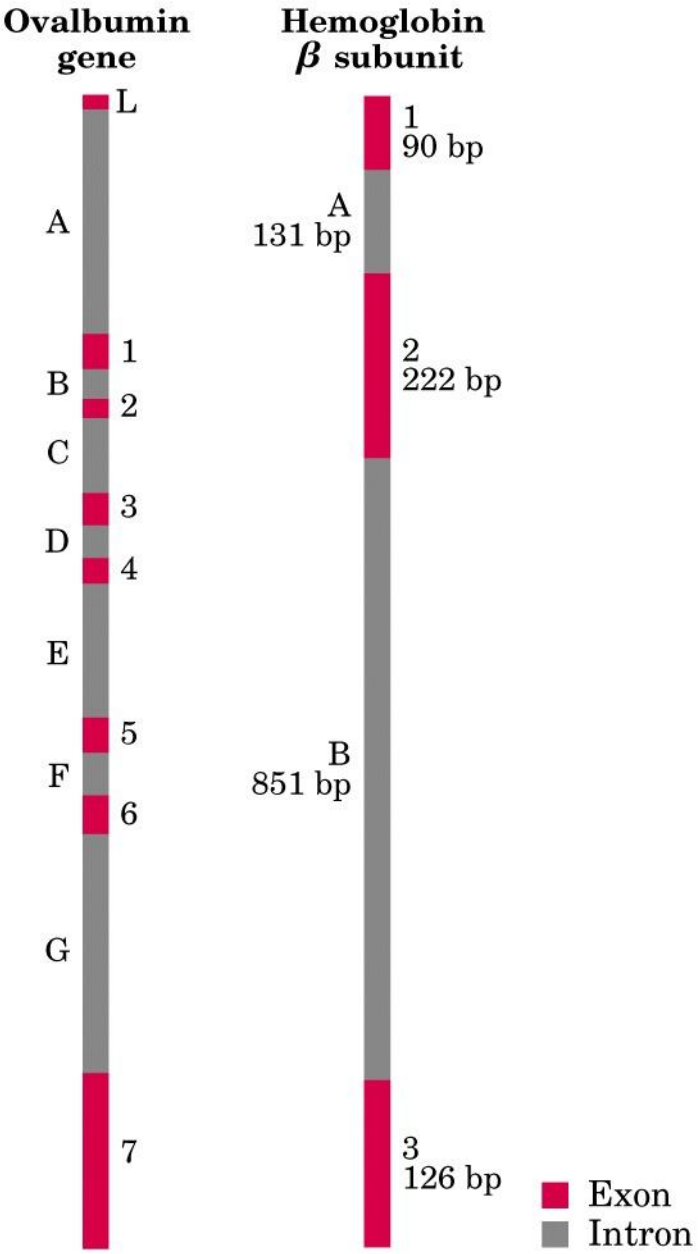
Both exons and introns are transcribed to yield a long primary RNA transcript.

The introns are removed by splicing to form the mature mRNA.

Mature mRNA is exported to cytoplasm for translation by ribosomes:

Proteins
RNA

Ovoalbumin gene: 7 introns A-G
Hemoglobin β gene: 2 introns A y B



Defining functional DNA elements in the human genome

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With the completion of the human genome sequence, attention turned to identifying and annotating its functional DNA elements. As a complement to genetic and comparative genomics approaches, the Encyclopedia of DNA Elements Project was launched to contribute maps of RNA transcripts, transcriptional regulator binding sites, and chromatin states in many cell types. The resulting genome-wide data reveal sites of biochemical activity with high positional resolution and cell type specificity that facilitate studies of gene regulation and interpretation of noncoding variants associated with human disease. However, the biochemically active regions cover a much larger fraction of the genome than do evolutionarily conserved regions, raising the question of whether nonconserved but biochemically active regions are truly functional. Here, we review the strengths and limitations of biochemical, evolutionary, and genetic approaches for defining functional DNA segments, potential sources for the observed differences in estimated genomic coverage, and the biological implications of these discrepancies. We also analyze the relationship between signal intensity, genomic coverage, and evolutionary conservation. Our results reinforce the principle that each approach provides complementary information and that we need to use combinations of all three to elucidate genome function in human biology and disease.

Definition of a Gene

“There are three aspects to the definition that we will list below, before providing the succinct definition:

1. A gene is a genomic sequence (DNA or RNA) directly encoding functional product molecules, either RNA or protein.

2. In the case that there are several functional products sharing overlapping regions, one takes the union of all overlapping genomic sequences coding for them.

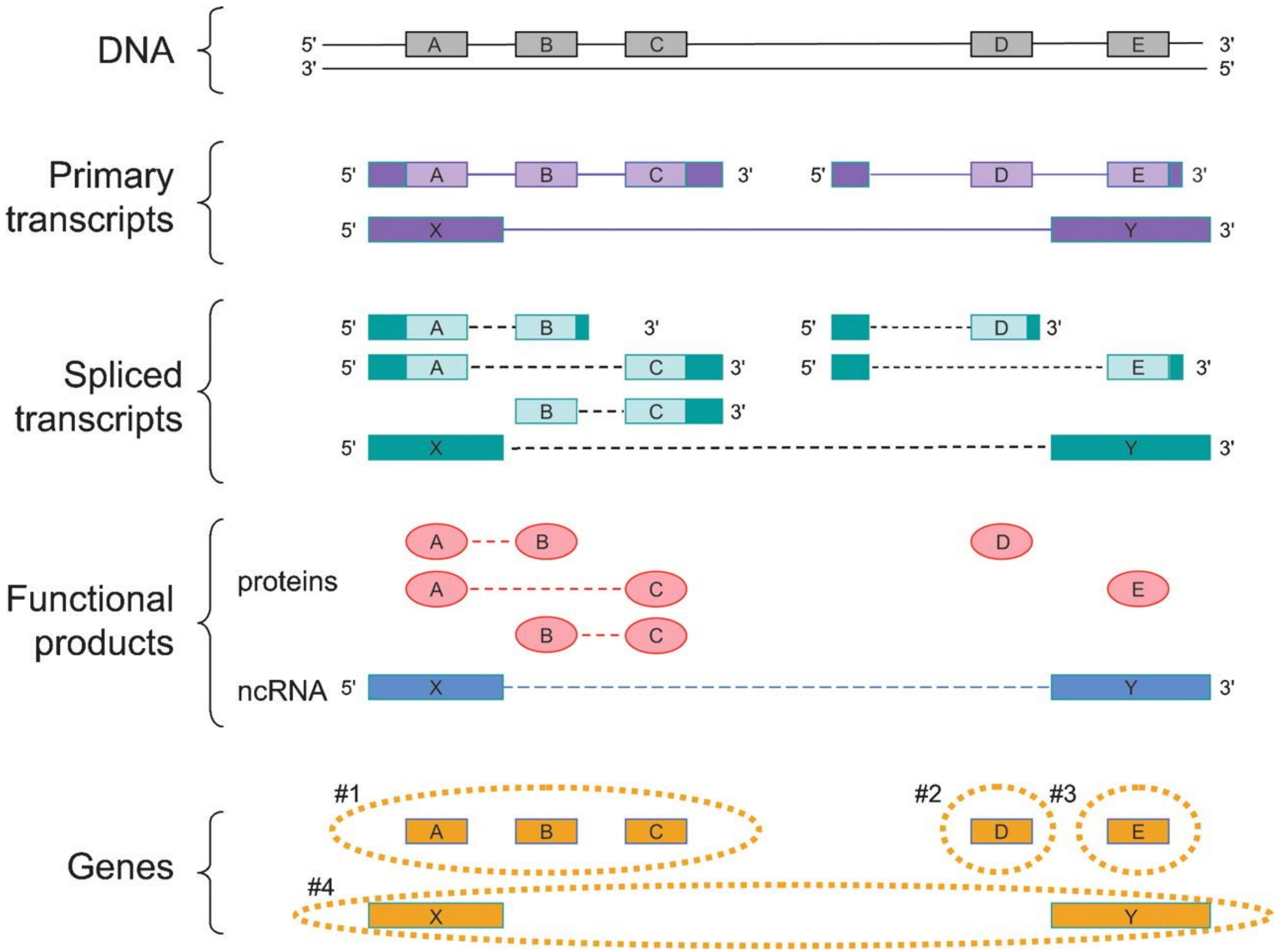
3. This union must be coherent—i.e., done separately for final protein and RNA products—but does not require that all products necessarily share a common subsequence.

This can be concisely summarized as:

The gene is a union of genomic sequences encoding a coherent set of potentially overlapping functional products.”

Gerstein, M. B., Bruce, C., Rozowsky, J. S., Zheng, D., Du, J., Korbel, J. O., ... & Snyder, M. (2007). What is a gene, post-ENCODE? History and updated definition. *Genome research*, *17*(6), 669-681.

Figure 5. How the proposed definition of the gene can be applied to a sample case



Mark B. Gerstein et al. Genome Res. 2007; 17: 669-681

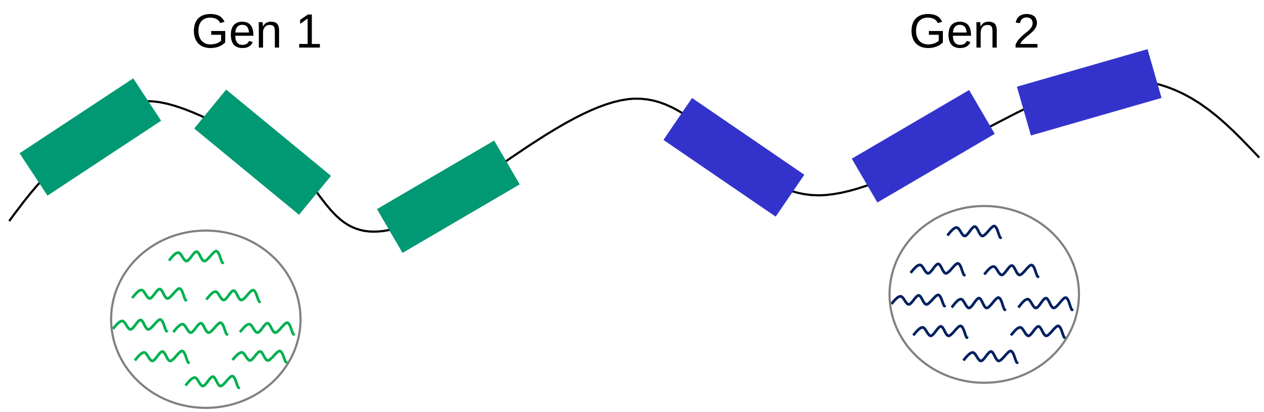
Cold Spring Harbor Laboratory Press

GENES IN THE HUMAN GENOME

The human genome contains about 12% single exonic genes (Sakharkar et al. 2004).

gene	size (kb)	number of exons	Average exon size (bp)	Average intron size (bp)
Histone H4	0.4	1	300	-
tRNA	0.1	2	50	20
insulin	1.4	3	155	480
β-globin	1.6	3	150	490
Class I HLA	3.5	8	187	260
serum albumin	18	14	137	1,100
type VII collagen	31	118	77	90
complement C3	41	29	122	900
factor VIII	186	26	375	7,100
CFTR	250	27	227	9,100
Dystrophin	2400	79	180	30,000

Gene annotation



Genome annotation is the process of deriving the structural and functional information of a protein or gene from a raw data set using different analysis.

Transcriptome

DNA Sequencing

cDNA Complementary DNA
EST Expressed Sequence Tags
SAGE Serial Analysis Gene Expression
CAG CAP analysis Gene Expression

RNA Sequencing

Total RNA
mRNA
Small RNA
mi RNA

Annotation

Biochemical

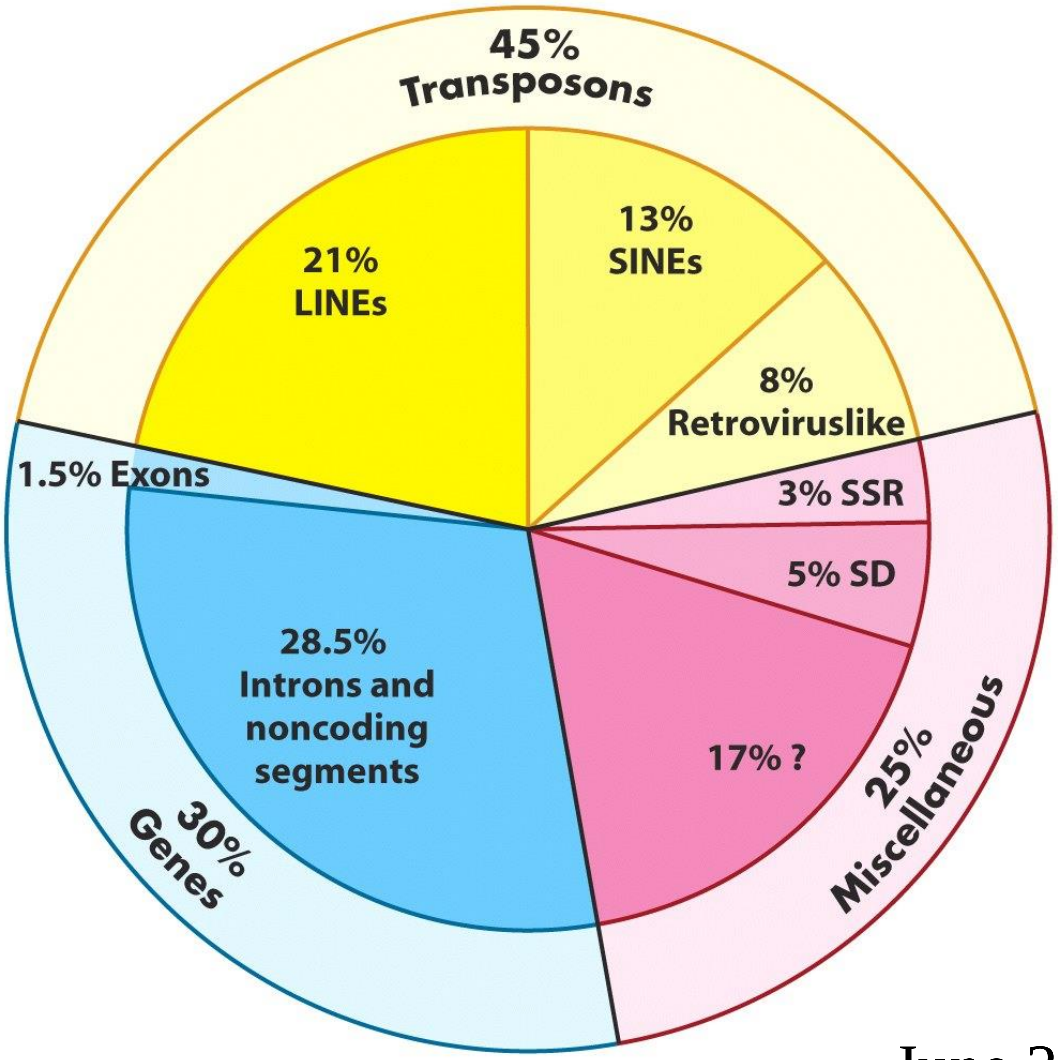
Genetics

Protein function (coding)

Evolution

DNA ELEMENTS OF THE HUMAN GENOME

- 3,000, 0000,0000 bp (3 Gbp, 3 x 10^9 bp)



June 2000/March 2022

DNA Elements in the Human Genome

Tandem repetitive DNA sequences

Tandem repetitive DNA sequences are **short tracts of DNA that are repeated multiple times in the genome.** These sequences are also called VNTRs, or variable number tandem repeats, SSRs, simple sequence repeats. Abundance, 8% of human genome sequence.

These elements are highly polymorphic as individuals within a population may have different numbers of repeats.

Three major classes:

- Satellite
- Minisatellite
- Microsatellite

DNA Elements in the Human Genome

Tandem repetitive DNA sequences

Satellite DNA

Highly repetitive DNA consisting of short sequence units repeated a large number of time forming lon arrays (up to 100 Mb).
Satellites are located in the heterochromatin regions of chromosomes (centromeres and subtelomeres) and are transcriptionally silent.

The repetitive unit (monomer) ranges from 150 to 400 bp.

Human centromeres are located at regions of A-T rich ɑ satellite.

ɑ satellite DNA is composed of 171bp monomers that are 50–70% identical in the array. It is the main DNA element of the human centromere and pericentromeric regions. (Schueler et al., 2001).

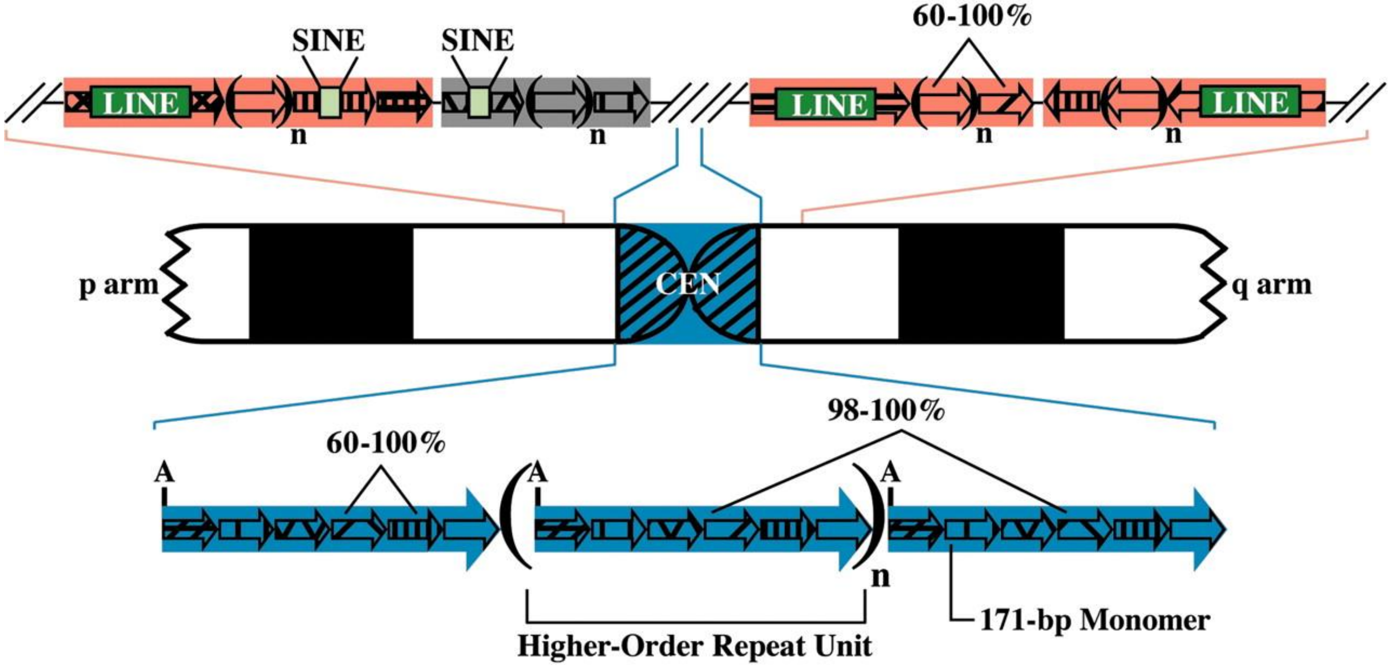
SatDNA is very variable in length between individuals.

Sullivan LL, Chew K, Sullivan BA. ɑ satellite DNA variation and function of the human centromere. Nucleus. 2017 Jul 4;8(4):331-339.

Centromeres

- Centromeres are formed by tandem repetitive DNA sequence located in all human chromosomes.
- Human centromeres are among the most diverse and rapidly evolving regions of the genome.
- Human centromeric DNA is tipically composed of tandemly repeating, approximately 171 bp α-satellite DNA, which is organized into HOR (High-order repetitive) units that can extend for megabase pairs (Mb) of sequence.
- Centromeres are essential chromosomal structures for chromosome segregation during cell division (both mitosis and meiosis).
- Centromeres are highly specialized genomic structure that function during cell division to ensure accurate chromosome segregation and maintain genome stability.
- Centromeres are the primary constriction of metaphase chromosomes where they serve as the assembly site for the kinetochore, a multiprotein structure that attach to the microtubules of the mitotic and meiotic spindles.

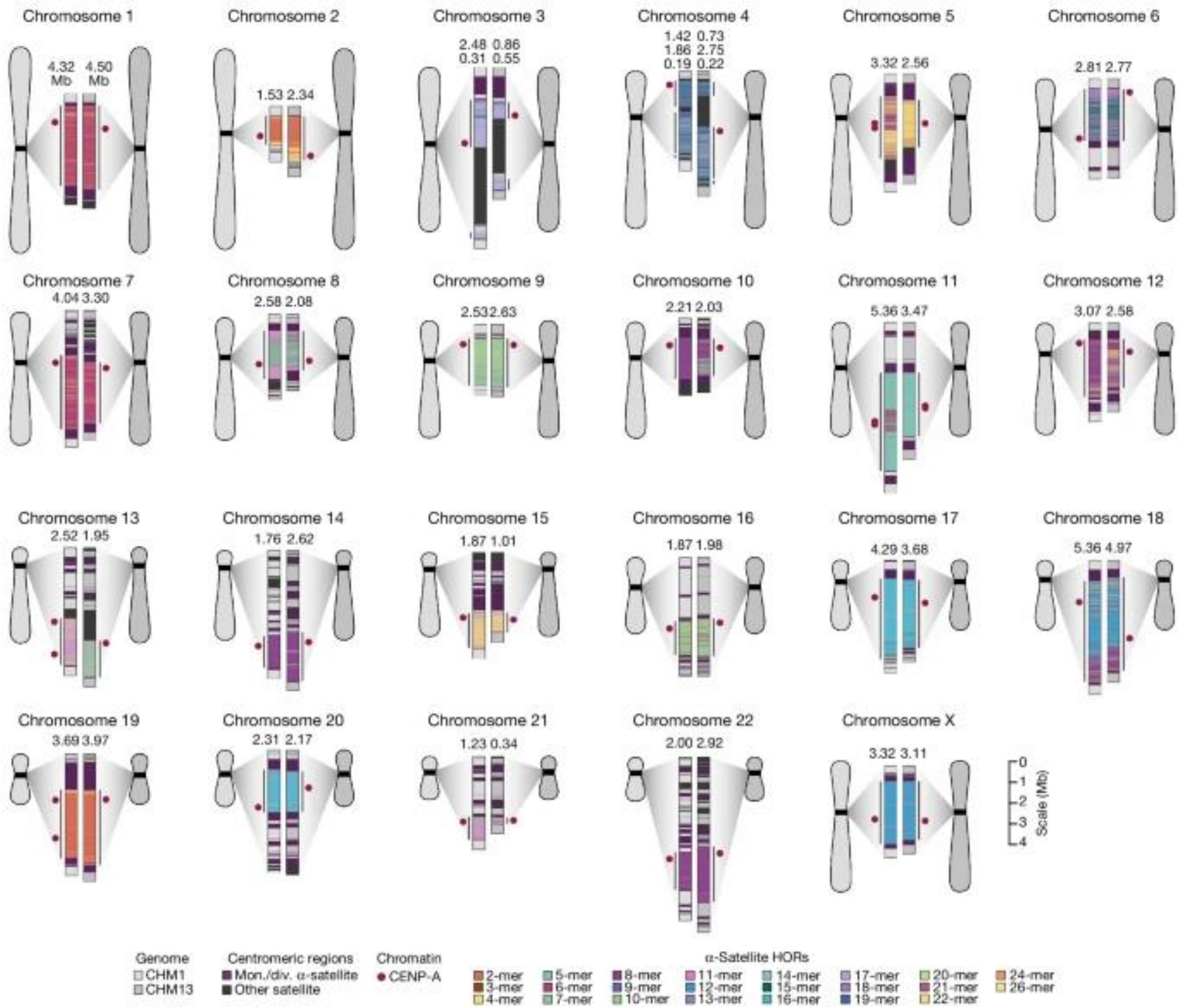
Fig. 1. Structural organization of human centromeric and pericentromeric regions



Schueler, Mary G. et al. (2005) Proc. Natl. Acad. Sci. USA 102, 10563-10568

PNAS

Centromere Variation



DNA Elements in the Human Genome

Tandem repetitive DNA sequences

Minisatellite DNA (VNTRs, variable number of tandem repeats)

- Minisatellites are tracts of tandem repetitive DNA with a DNA repeat unit ranging from 10 to 60 bp that tipically span to several kb of length.
- Highly polymorphic due to high mutation rates. Thus, humans have a high diversity in minisatellite sequence.
- Minisatellites are located in the subtelomere, telomere an centomere of chromosomes, but they are also found in different locations in chromosomes.
- They do not encode protein or other functional element so far.
- They ae used in forensics: DNA profiling.

Table 1. Known Human Hypermutable Minisatellites

Minisatellite name	Locus or D-number	Location	Accession number	Consensus length	Nb of variant positions	Percent match (%M)	GC% (bias)	Purine % (bias)	Instability	Proportion of paternal events
CEB1	D2S90	2q37.3 (T)	AF048727	39	8	90	72 (0.53)	68 (0.38)	6.7%	97%
CEB15	D1S172	1p36.33 (T)	AL096805	18	8	91	68 (0.33)	77 (0.56)	1.5%	100%
CEB25	D10S180	10q26.3 (T)	AL096806	52	>20	82	53 (0.73)	67 (0.36)	2.5%	65%
CEB36	D10S473	10q26.3 (T)	AL096810	42	18	93	64 (0.65)	66 (0.32)	1.8%	50%
CEB42	D8S358	8q24.3 (T)	AL096807	41	4	95	63 (0.11)	53 (0.08)	0.5%	100%
CEB72	D17S888	17q25 (T)	AL096808	21	0	100	71 (0.60)	80 (0.62)	1.8%	65%
MS1	D1S7	1p33-35 (I)	X14856	9	3	85	58 (0.93)	79 (0.60)	5%	50%
MS32	D1S8	1q42 (I)	AF048729	29	3	92	61 (0.37)	67 (0.35)	0.4%	50%
B6.7		20q13.3 (T)	AF081787	34	6	91	60 (0.5)	68 (0.37)	5.5%	64%
CSTB	EPM1	21q22.3 (T)	U46692	12	1	96	100 (0.5)	100 (0.5)	47%	76%

Minisatellites showing instability higher than 1% in the male or female germline are listed (with the exception of MS32, with a lower mutation rate, but which is a reference minisatellite in many investigations). The instability values indicated are most often average values, as measured usually in the large Centre d'Etudes du Polymorphisme Humain families (http://www.cephb.fr) with the exception of the cystatin B (CSTB) gene minisatellite. In this last case, the values given were measured in pathogenic, expanded alleles (Larson et al. 1999). Similarly, the mutation rate at CEB1 alleles has been shown to vary between <0.02% (at smaller alleles) and >20% (Buard et al. 1998). Percent match (%M) is the average similarity of motifs with the consensus motif. GC bias is the absolute value of (G% − C%)/(G% + C%). Purine bias is the absolute value of (pur% − pyr%).

DNA Elements in the Human Genome

Tandem repetitive DNA sequences

Microsatellite DNA (STR, short tandem repeats)

Microsatellite are short tracts of tandem repetitive DNA with repeat unit ranging from 1 to 13 bp spanning up to 500 bp length.

A microsatellite have a mono-, di-, tri-, or tetranucleotide unit repeat (penta- and hexa- are also reported)

They are analysed by simple PCR reactions and DNA sequencing.

Used in parentage analysis. Paternity determination (12 microsatellites)

In humans, microsatellites in the form of trinucleotide repeats can be found in genes that are associated with several neurological diseases.

The association of triplet repeats with genes provides a basis for identifying genes that may be predisposed to expansion.

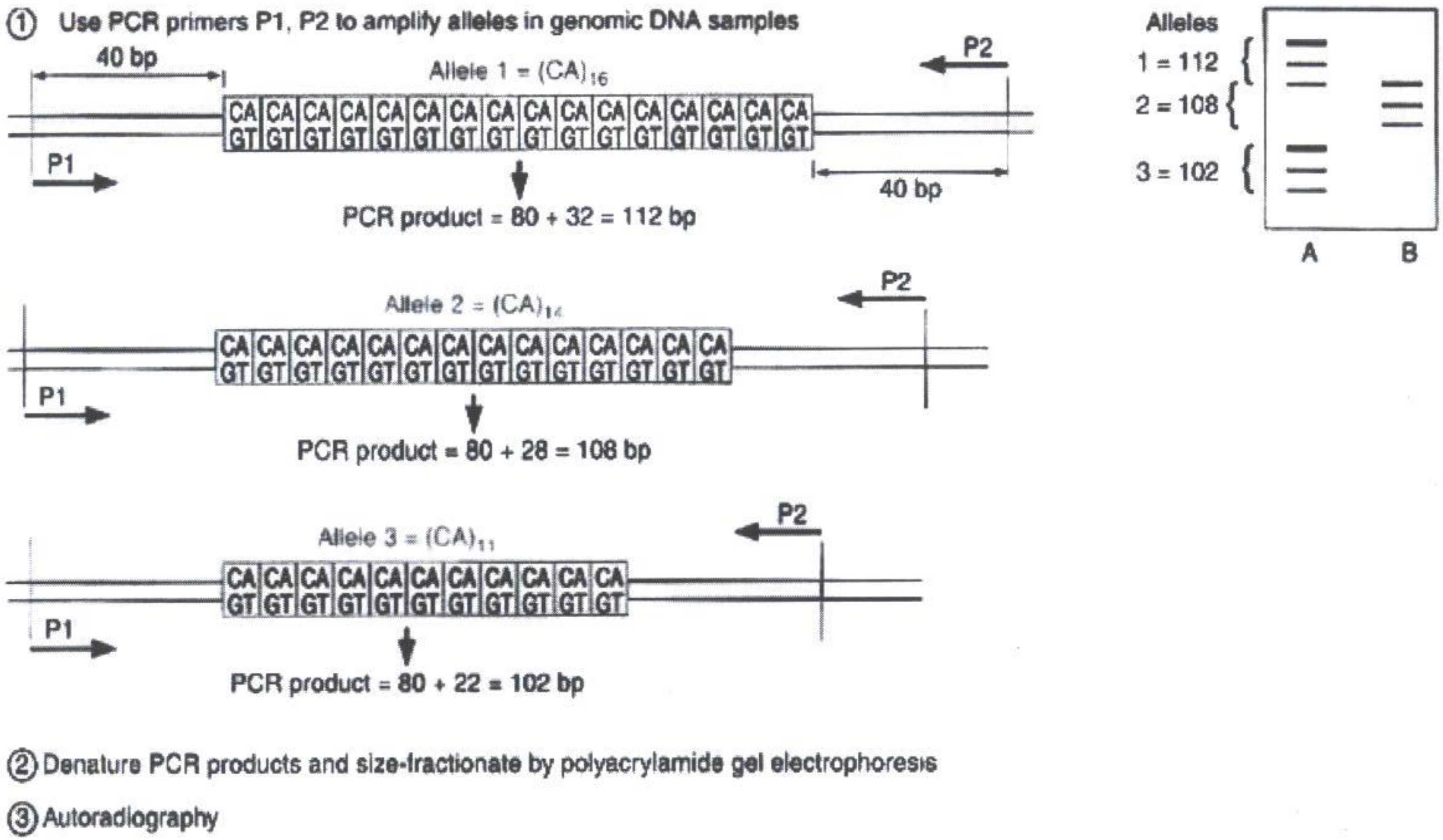


Figura 2. Uso de PCR en análisis de Microsatélites (STR)

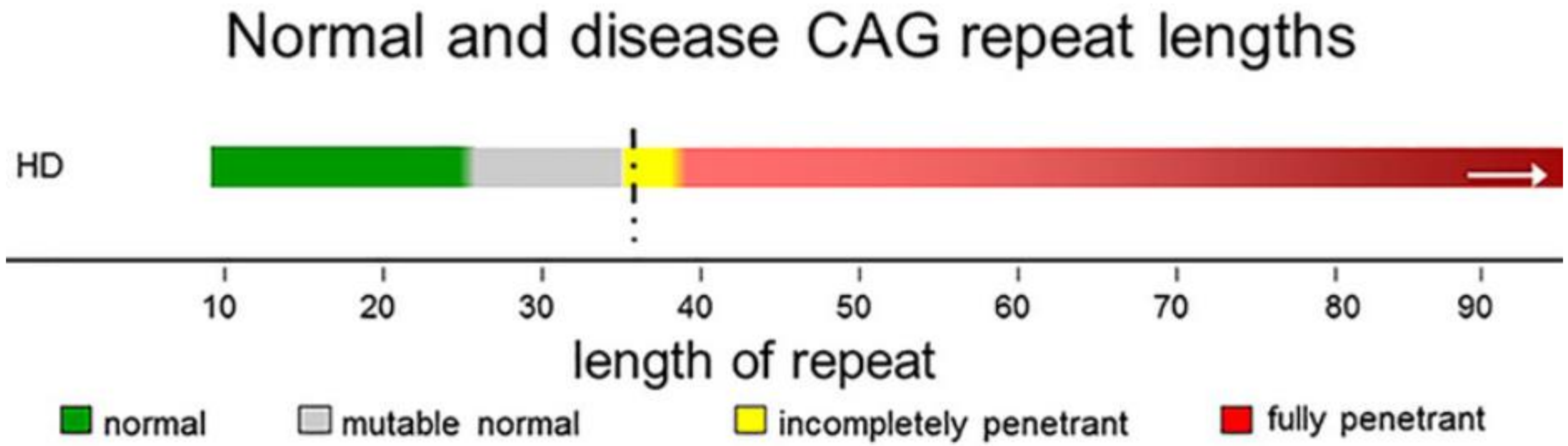
Microsatellite expansion and neurological diseases

1. Very large expansions of repeats outside coding sequences						
Fragile-X site A (FRAXA)	309550	X	Xq27.3	5'UT	$(CGG)_n$	6–54
Fragile-X site E (FRAXE)	309548	X	Xq28	Promoter	$(CCG)_n$	6–25
Friedreich ataxia (FA)	229300	AR	9q13-q21.1	Intron 1	$(GAA)_n$	7–22
Myotonic dystrophy (DM)	160900	AD	19q13	3'UT	$(CTG)_n$	5–35
Spinocerebellar ataxia 8	-	AD	13q21	Untranslated RNA	$(CTG)_n$	16–37
Juvenile myoclonus epilepsy (JME)	254800	AR	21q22.3	Promoter	$(CCCCGCCCCGCG)_n$	2–3
2. Modest expansions of CAG repeats within coding sequences						
Huntington disease (HD)	143100	AD	4p16.3	Coding	$(CAG)_n$	6–35
Kennedy disease (SBMA)	313200	XR	Xq21	Coding	$(CAG)_n$	9–35
Spinocerebellar ataxia 1 (SCA1)	164400	AD	6p23	Coding	$(CAG)_n$	6–38
Spinocerebellar ataxia 2 (SCA2)	183090	AD	12q24	Coding	$(CAG)_n$	14–31
Machado-Joseph disease (SCA3, MJD)	109150	AD	14q32.1	Coding	$(CAG)_n$	12–39

Huntington Disease

The huntingtin (HTT) gene has 67 exons spanning 180 kb of genomic DNA, that is approximately 3144 amino acids ~ 348 kDa.

HTT has a polyglutamine repeat (translated from the CAG repeat) at the N-terminus.



https://www.youtube.com/watch?v=qTVTLVd0P5o

Satellite DNA	Repeat Unit	Location
Length 100 kb to Mb		
α Satellite DNA	171 bp	Heterochromatin
β Satellite DNA	68 bp	Heterochromatin chromosomes 13, 14, 15, 21, 22
Satellite 1	25-48 bp	Centromeres and other regions in heterochromatin of most chromosomes
Satellite 2	5 bp	Most chromosomes
Satellite 3	5 bp	Most chromosomes
Minisatellite	**Repeat Unit**	**Location**
Length 1 kb to 20 kb (or more)		
Telomere	6 bp	All chromosomes
Hypervariable family	9-24 bp	Subtelomere
Microsatellite	**Repeat Unit**	**Location**
Length less than ≈500 bp	1-5 bp	All chromosomes

DNA Elements in the Human Genome

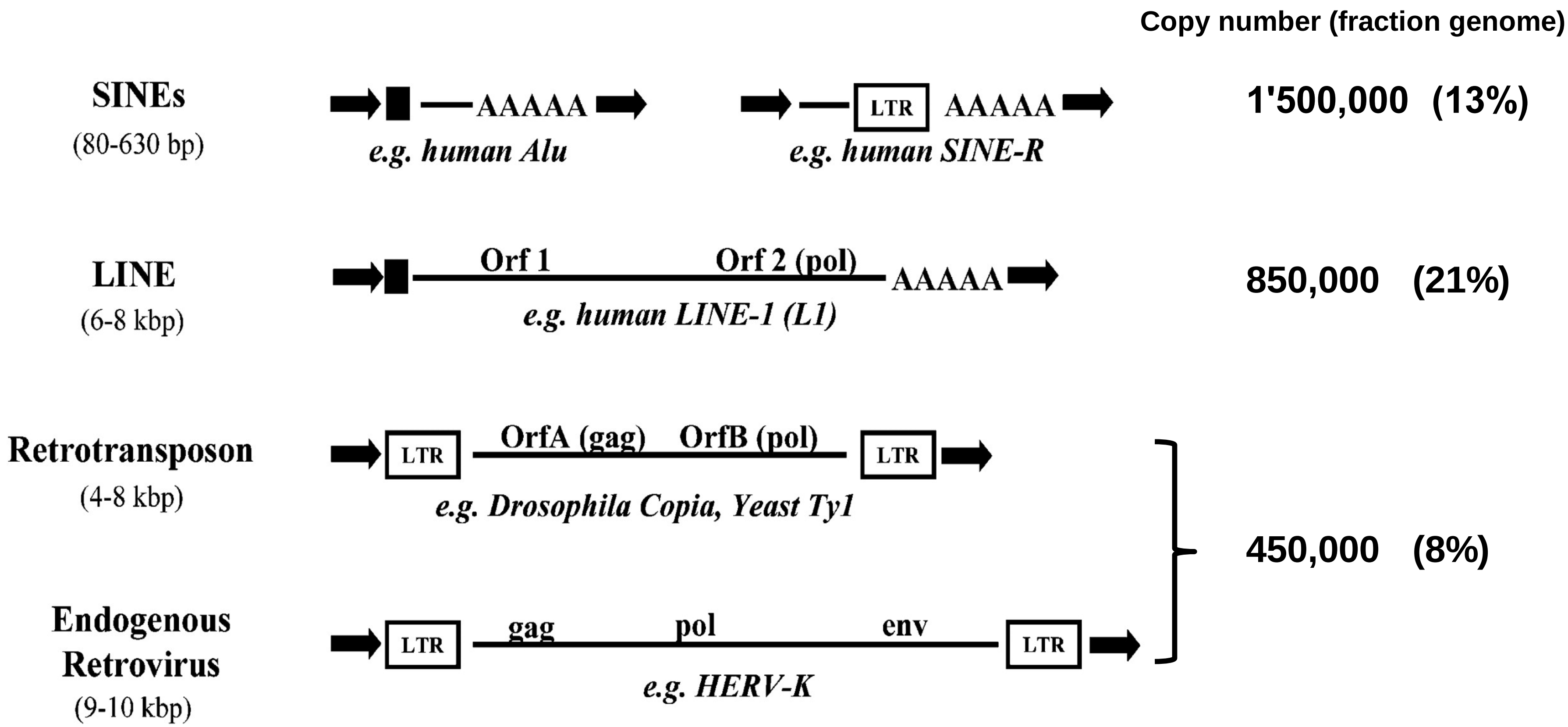
MOBILE ELEMENTS

Transposable Elements, ~45%; $3x10^6$

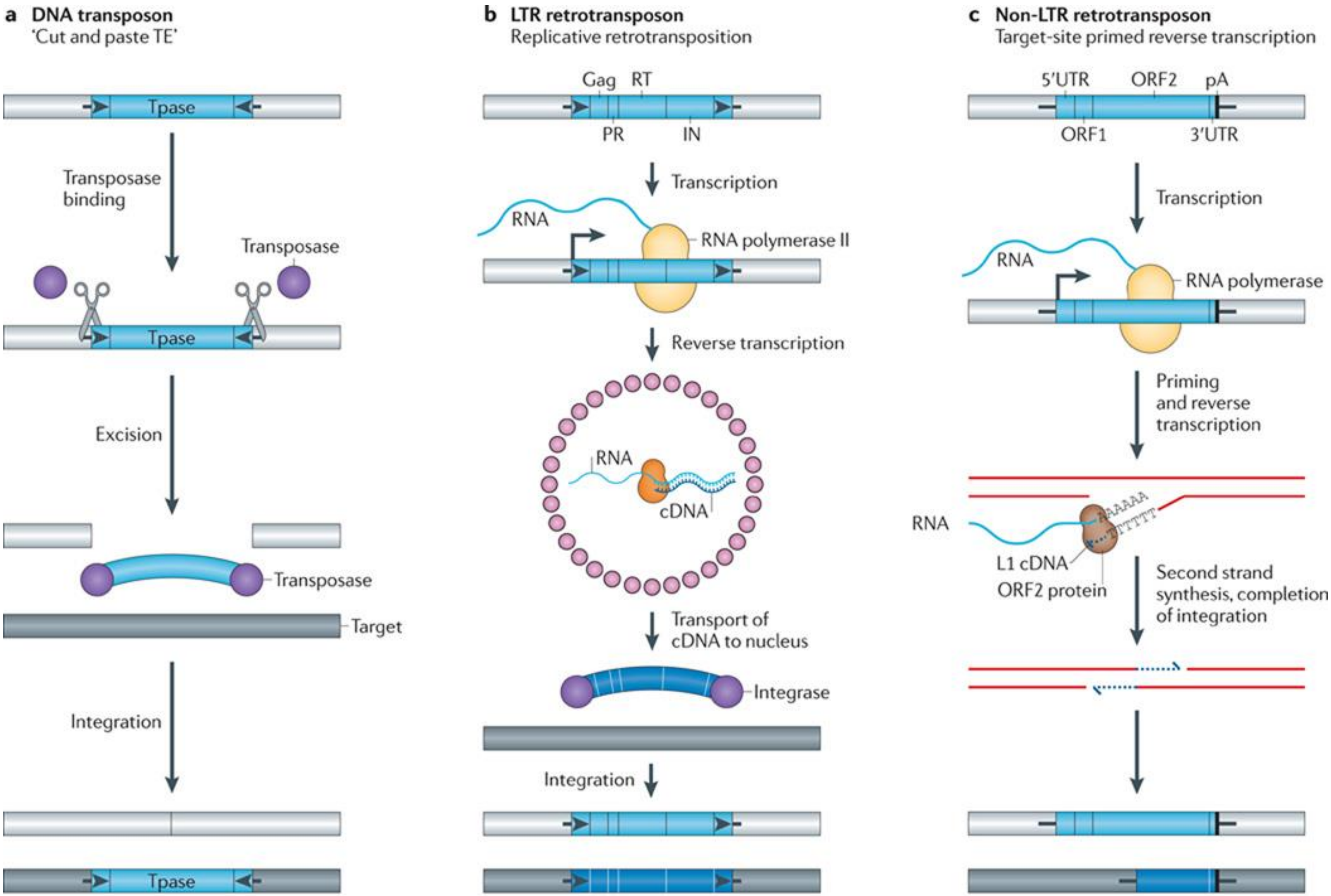
- Transposons, 2.8%; $0.3x10^6$
- RNA intermediate: Retroelements, 42.2%; $2.7x10^6$
  - non LTR, 33.9%; $2.4x10^6$
    - SINE (Alu, 10.6%; MIR, 2.5%)
    - LINE (L1, 16.9%; L2, 3.2%)
    - processed pseudogenes, <1.0%
  - LTR, 8.3%; $0.3x10^6$
    - class I ERV + MER4, 2.8%
    - class II ERV, 0.3%
    - class III ERV, 4.6%
    - others (MST, MLT, etc.), 0.6%

Bannert, Norbert and Kurth, Reinhard (2004) Proc. Natl. Acad. Sci. USA 101, 14572-14579

Abundance and structural features of important retroelements in human genome



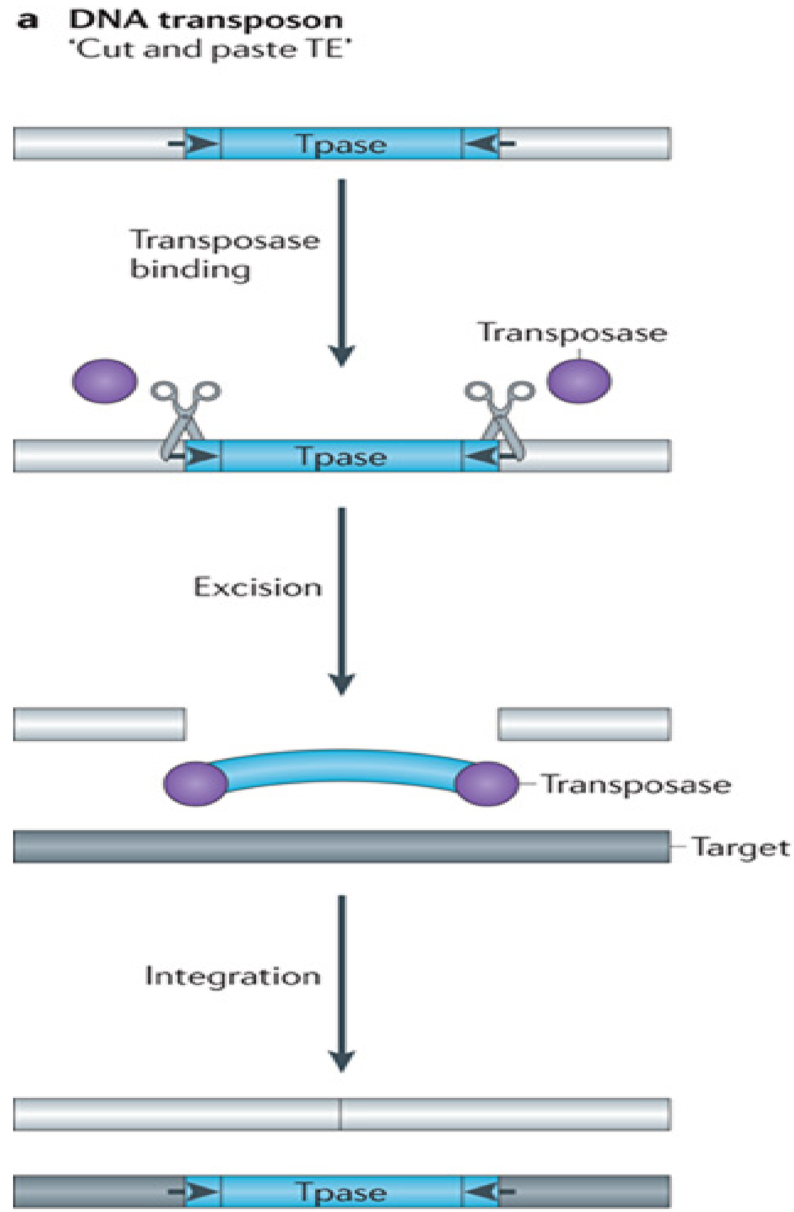
Bannert, Norbert and Kurth, Reinhard (2004) Proc. Natl. Acad. Sci. USA 101, 14572-14579



Nature Reviews | Genetics

DNA transposon

- Make up ~3% of the human genome.
- “no evidence for DNA transposon activity in the past 50 My in the human genome” (Lander et al. 2001)
- DNA transposon are inactive DNA elements and are considered “ancient genomic relics”, no capable of mobilizing.
- Many DNA transposons are flanked by terminal inverted repeats (TIRs, black arrows).
- Encode a transposase, and mobilize by a “cut and paste” mechanism.
- Transposase binds at or near the TIRs, excises the transposon from its existing genomic location (light gray bar), and pastes it into a new genomic location (dark gray bar).
- Cleavage of the two strands at the target site are staggered, resulting in a target site duplication (TSD), typically 4 to 8 bp as specified by the TPase (Short black lines flanking the TE).



Long terminal repeat transposons

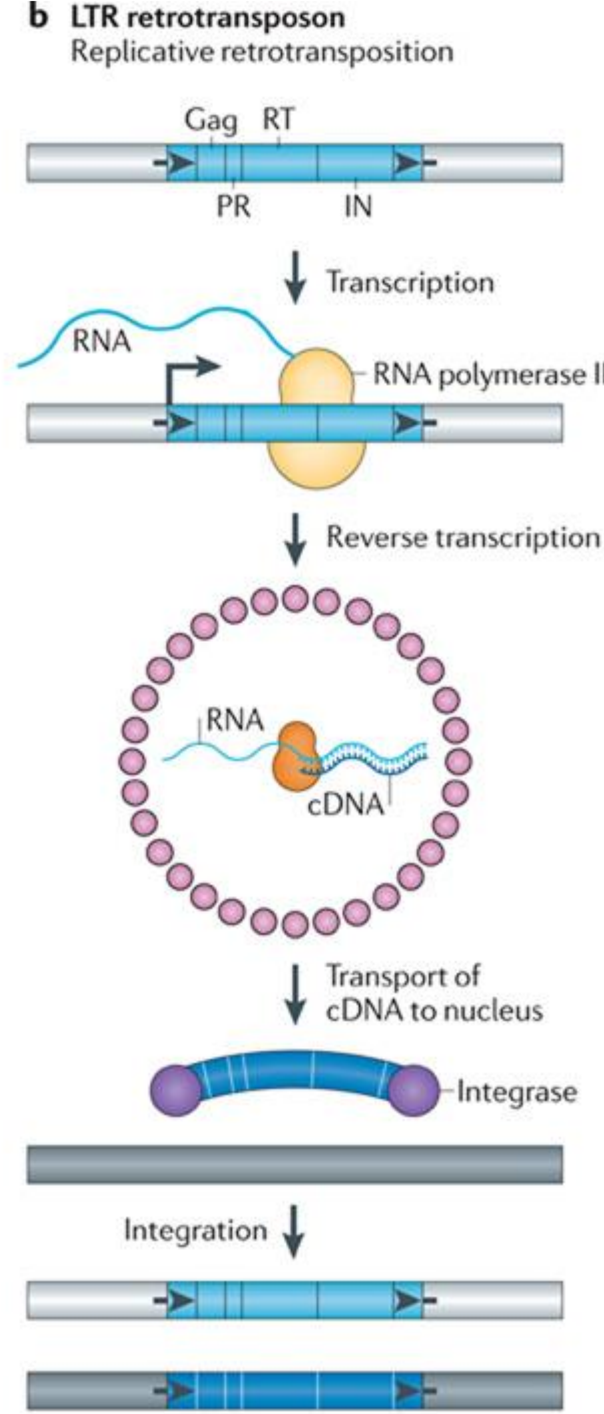
Retrotransposons mobilize by a replicative mechanism that requires the reverse transcription of an RNA intermediate.

LTR-retrotransposons contain two long terminal repeats (LTRs, black arrows) and encode Gag, protease (PR), reverse transcriptase (RT), and integrase (IN) activities critical for retrotransposition.

The 5′ LTR contains a promoter of host RNA polymerase II and produces the mRNA of the TE (start of transcription a black vertical line with a right facing arrow).

Step 1, Gag (small pink circles) assembles into virus like particles containing TE mRNA, RT, and IN. The RT reto-transcribe TE mRNA into a full-length double stranded DNA (wide blue arc).

Step 2, IN (purple circles) inserts the DNA into the new target site. Similar to the TPases of DNA transposons, INs create staggered cuts at the target sites that result in integraton of the LTR-Transposon.



Non-LTR retrotransposons (LINE and SINE)

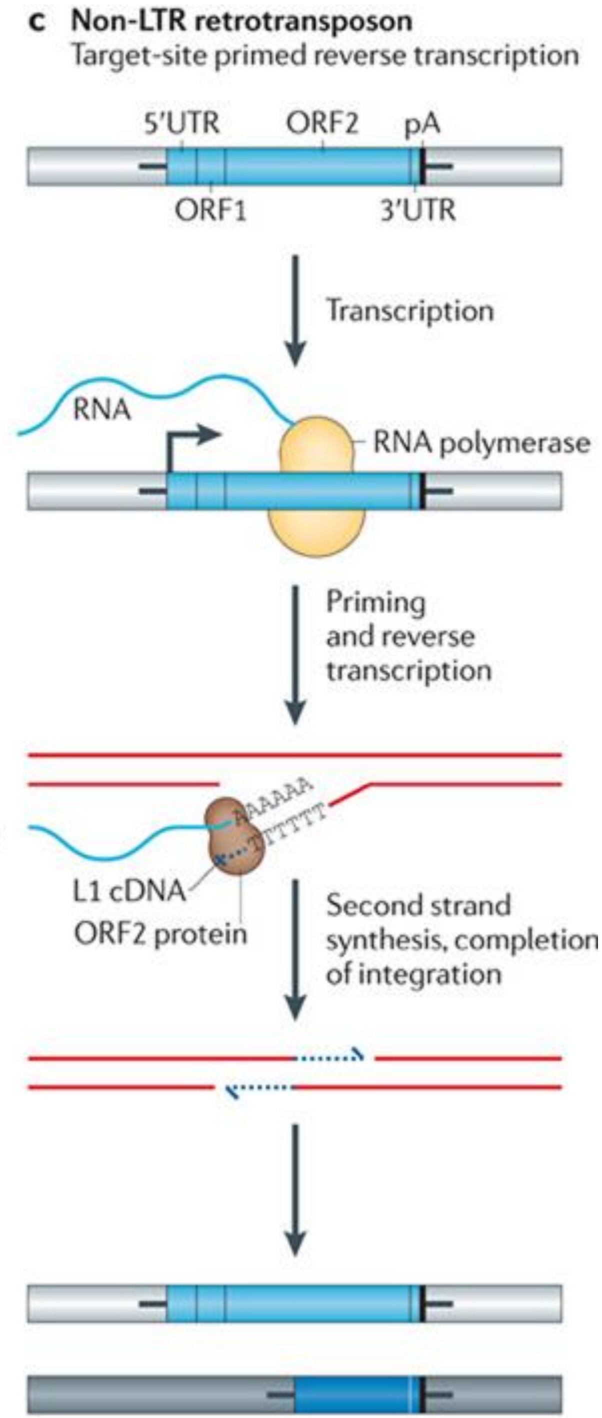
Lack LTRs and encode either one or two open reading frames.

The transcription of non-LTR retrotransposons produces a full-length mRNA.

These elements mobilize by target-site primed reverse transcription (TPRT): An endonuclease generates a single-strand “nick” in genomic DNA, liberating a 3’OH that is used to prime reverse transcription of the RNA.

The TPRT mechanism of an L1 element.

The integration of non-LTR retrotransposons can lead to target-site duplications (TSDs) and small deletions at the target site in genomic DNA.



Mobile elements insertions and disease

Table 1. Pathologic Intronic Insertions in Humans

Retroelement	Length	Orientation	Position in Gene	Gene Name	Abbreviation	Phenotype	mRNA Studies	Reference
Alu								
*Alu*Yc1	316 bp	antisense	−52 bp from the 3′ end of intron 4	glycerol kinase	*GK*	benign isolated glycerol kinase deficiency	not reported	(Zhang et al., 2000)
*Alu*Yb9	~330 bp	antisense	−19 bp from the 3′ end of intron 18	coagulation factor VIII	*F8*	hemophilia A	exon 19 skipping	(Ganguly et al., 2003)
*Alu*Ya5	331 bp	antisense	−50 bp from the 3′ end of intron 7	tumor necrosis factor receptor superfamily, member 6	*FAS*	autoimmune lymphoproliferative syndrome	exon 8 skipping	(Tighe et al., 2002)
*Alu*Ya5	368 bp	antisense	−19 bp from the 3′ end of intron 8	fibroblast growth factor receptor 2	*FGFR2*	Apert syndrome	ectopic exon 7/8 splicing in lieu of 7/9	(Oldridge et al., 1999)
*Alu*Ya5	320 bp	antisense	−44 bp from the 3′ end of intron 5	neurofibromatosis type 1	*NF1*	neurofibromatosis type 1	exon 6 skipping	(Wallace et al., 1991)
L1								
L1(Ta)	836 bp	sense and rearranged	intron 5	cytochrome b-245, β polypeptide	*CYBB*	chronic granulomatous disease (CGD)	variable L1 exonization; exon 5 and exon 6 skipping	(Meischl et al., 2000)
L1(Ta)	6 kb	antisense	reported as 3′ end of intron 2	hemoglobin, β	*HBB*	β-thalasemia	not reported	(Kimberland et al., 1999)
L1(Ta)	1.2 kb	sense	−24 bp from the 3′ end of intron 7	fukutin	*FKTN*	Fukuyama-type congenital muscular dystrophy	variable exon 7, 8, and 9 skipping	(Kondo-Iida et al., 1999)
L1(Ta)	6 kb	sense	intron 1	retinitis pigmentosa 2	*RP2*	X-linked retinitis pigmentosa	no transcript detected by RT-PCR	(Schwahn et al., 1998)
L1(Ta)	2.8 kb	antisense and rearranged	−8 bp from the 3′ end of intron 3	ribosomal S6 kinase 2 gene	*RPS6KA3*	Coffin-Lowry syndrome	exon 4 skipping	(Martinez-Garay et al., 2003)

HUMAN ENDOGENOUS RETROVIRUS (HERV)

- HERVs represent footprints of previous retroviral infection and can be separated into three classes according to the level of *pol* sequence homology.

- Classification is based on complementary binding between tRNA and the primer-binding site (PBS)

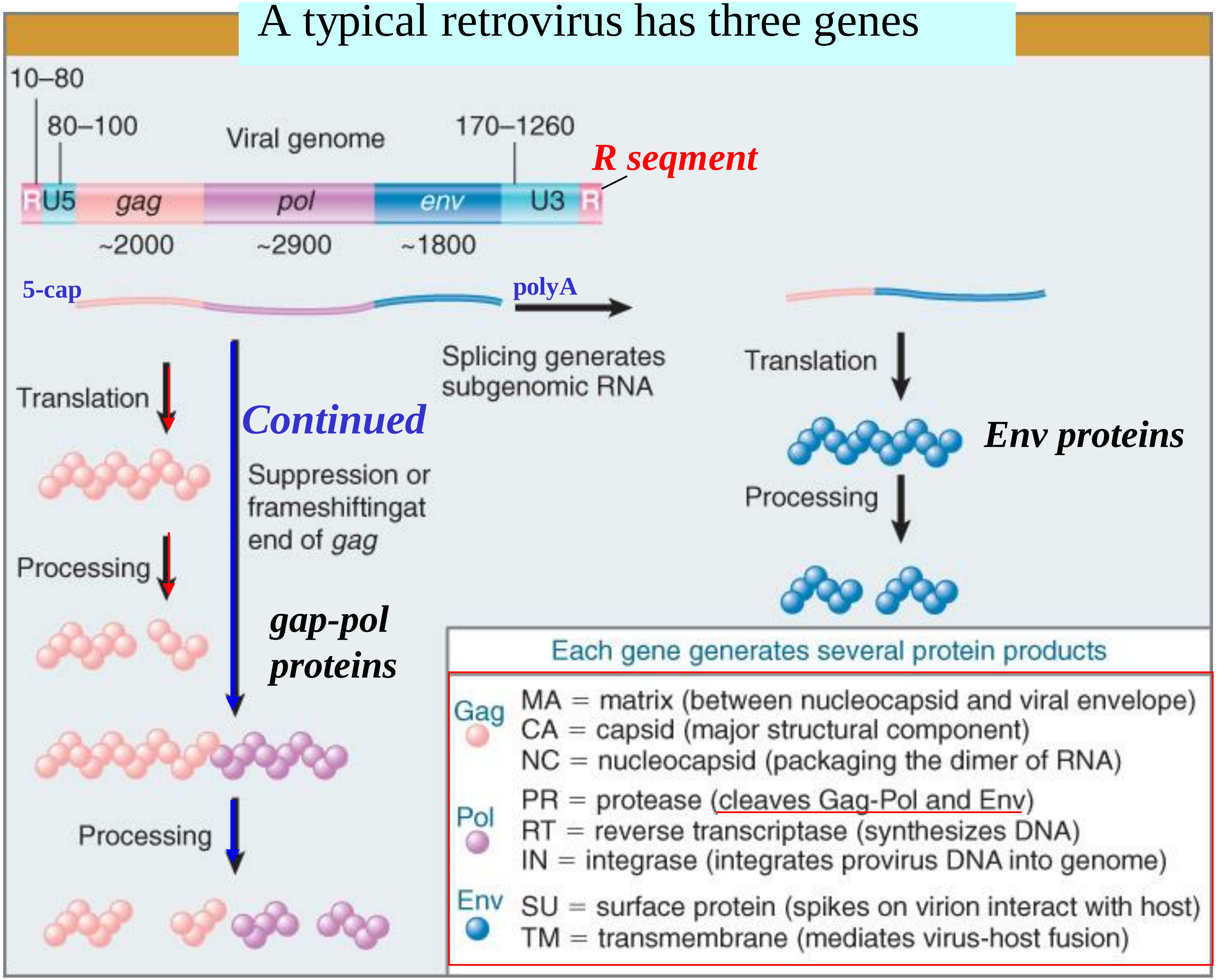
- A one-letter amino acid code is used to denote the corresponding HERV families: HERV-K, HERV-R, HERV-W, and HERV-H.

- The basic genes of HERVs are *gag, pro, pol*, and *env*.

  - *gag* gene codes: structural matrix (MA), capsid (CA), and nucleocapsid (NC) proteins.
  - pro codes for protease (PR) protein.
  - *pol* codes for the enzymes of reverse transcriptase (RT) and integrase (IN), and
  - *env* codes for the envelope surface (SU) and transmembrane (TM) proteins.

Kim, H. S. (2012). Genomic impact, chromosomal distribution and transcriptional regulation of HERV elements. *Molecules and cells*, *33*(6), 539-544.

HUMAN ENDOGENOUS RETROVIRUS



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ONCOLOGY

REVIEW ARTICLE
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HERV-K(HML-2), the best preserved family of HERVs: endogenization, expression, and implications in health and disease

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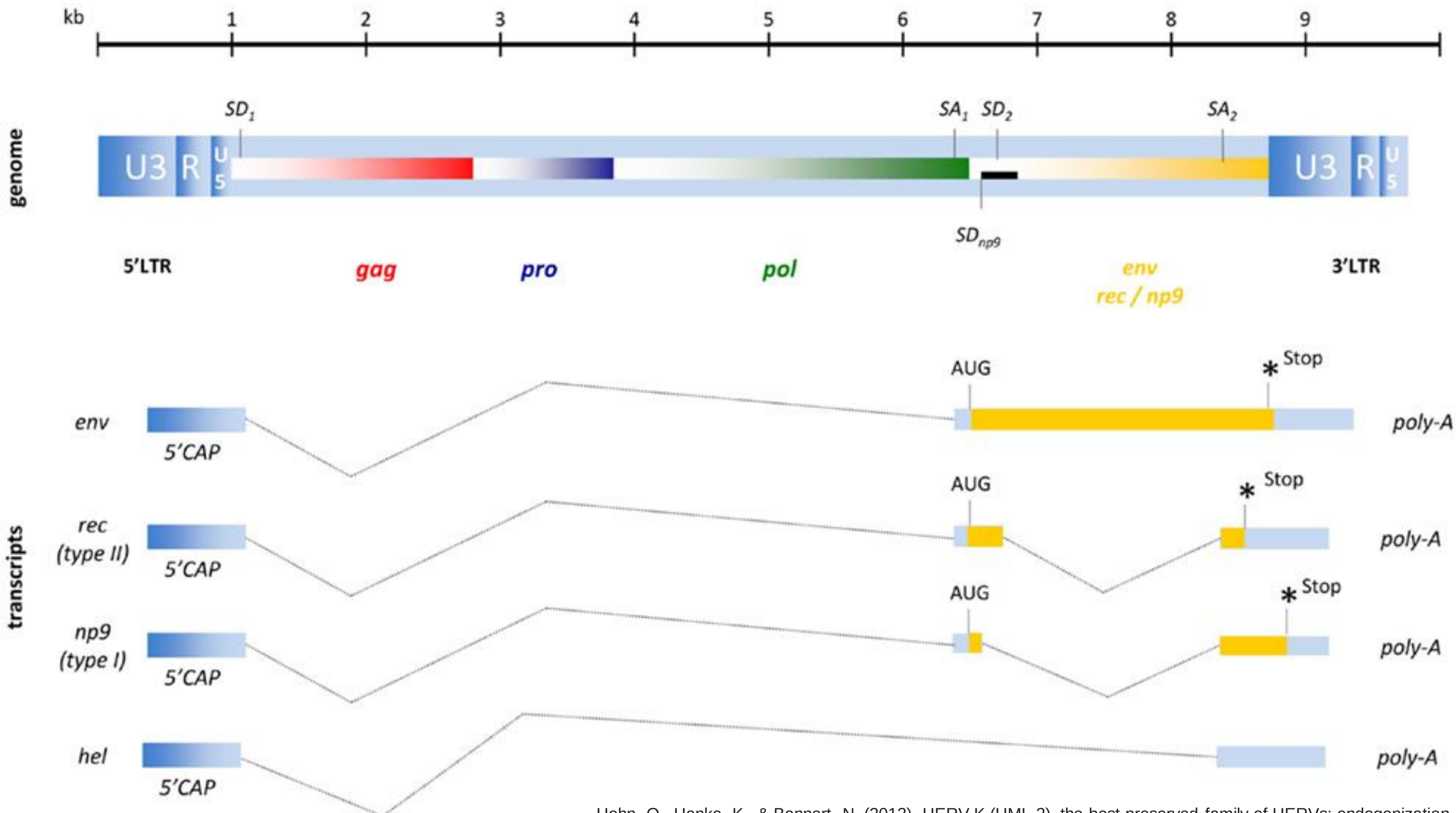
Retroviruses that have the ability to infect germ line cells can become an integral and inherited part of the host genome. About 8% of the human chromosomal DNA consists of sequences derived from infections by retroviruses that presumably circulated 2–40 millions of years ago, and some elements are actually much older. Post-insertional recombinations, deletions, and mutations have rendered all known human endogenous retroviruses (HERVs) non-infectious. However some, particularly the most recently acquired proviruses of the HERV-K(HML-2) family, can expresses viral proteins and produce viral particles. In this review we will first discuss the major aspects of the endogenization process and peculiarities of the different HERV-K families. We will then focus on the genes and proteins encoded by HERV-K(HML-2) as well as inactivation of these proviruses by postinsertional mutations and their inhibition by antiretroviral factors. After describing the evolutionary interplay between host and endogenous retrovirus we will delve deeper into the currently limited understanding of HERV-K and its possible association with disease, particularly tumorigenesis.

Keywords: human endogenous retrovirus, HERV-K, replication, Rec, cancer

Hohn, O., Hanke, K., & Bannert, N. (2013). HERV-K (HML-2), the best preserved family of HERVs: endogenization, expression, and implications in health and disease. *Frontiers in oncology*, *3*, 246.

Figure 2. Proviral organization of HERV-K(HML-2) and RNA transcripts. The black bar represents a 292-bp deletion that results in the Np9 gene product for type I proviruses instead of the Rec (an HIV-1 Rev homolog) characteristic of the type II proviruses. *Hel* is a poorly described RNA without protein coding function.



Hohn, O., Hanke, K., & Bannert, N. (2013). HERV-K (HML-2), the best preserved family of HERVs: endogenization, expression, and implications in health and disease. *Frontiers in oncology*, *3*, 246.

Human endogenous retrovirus-K(KML-2) and cancer

Table 2 | List of malignant diseases that are associated with HERV-K(HML-2) activity.

Tissue	Cancer	HERV-K(HML-2) activity	Reference
Skin	Melanoma	Retroviral particles Enhanced transcription RT activity Expression of Env, Rec, Np9	Buscher et al. (6), Muster et al. (86), Hirschl et al. (124)
Testes	Germ cell tumors, gonadoblastoma, seminoma	Anti-Gag/Env-Ab Expression of Rec, Np9	Boller et al. (87), Kleiman et al. (92), Boller et al. (125)
Ovary	Ovarian clear cell carcinoma; ovarian epithelial tumors	Expression of Gag and Env	Gotzinger et al. (15), Wang-Johanning et al. (72), Iramaneerat et al. (126)
Breast	Breast cancer	Free viral RNA RT activity Virus particles Specific CTLs	Wang-Johanning et al. (112), Contreras-Galindo et al. (127), Wang-Johanning et al. (128)
Prostate	Prostate cancer	Enhanced Gag-production due to fusion to androgen-dependent ETV1 and ETS genes	Tomlins et al. (12), Lamprecht et al. (13), Ishida et al. (111)
Blood	Lymphoma	Free RNA RT activity Virus-like particles	Contreras-Galindo et al. (127)

Hohn, O., Hanke, K., & Bannert, N. (2013). HERV-K (HML-2), the best preserved family of HERVs: endogenization, expression, and implications in health and disease. *Frontiers in oncology*, *3*, 246.

Table 3. Examples of retroviral sequences involved in the regulation of cellular proteins

Element	Gene (Promotor)	Functional Role	
HERV-E	Mid1	Opitz syndrome	76
HERV-E	Apolipoprotein C1	Liver and other tissues	78
HERV-E β	Endothelin-B receptor	Placenta	78
HERV-E	Pleiotrophin	Trophoblast	105
HERV-L	1,3-galactosyltransferase	Colon, mammary gland	106
ERV II	BAAT (transferase)	Bile metabolism	2
ERV I	Aromatase	Placental estrogen synthesis	2
ERV III	Carbonic anhydrase 1	Erythroid carbon metabolism	2
LTR + LINE-2	Chaperonin	McKausick–Kaufman syndrome	2

Retroelements and the human genome: New perspectives on an old relation

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Retroelements constitute a large portion of our genomes. One class of these elements, the human endogenous retroviruses (HERVs), is comprised of remnants of ancient exogenous retroviruses that have gained access to the germ line. After integration, most proviruses have been the subject of numerous amplifications and have suffered extensive deletions and mutations. Nevertheless, HERV-derived transcripts and proteins have been detected in healthy and diseased human tissues, and HERV-K, the youngest, most conserved family, is able to form virus-like particles. Although it is generally accepted that the integration of retroelements can cause significant harm by disrupting or disregulating essential genes, the role of HERV expression in the etiology of malignancies and autoimmune and neurologic diseases remains controversial. In recent years, striking evidence has accumulated indicating that some proviral sequences and HERV proteins might even serve the needs of the host and are therefore under positive selection. The remarkable progress in the analysis of host genomes has brought to light the significant impact of HERVs and other retroelements on genetic variation, genome evolution, and gene regulation.

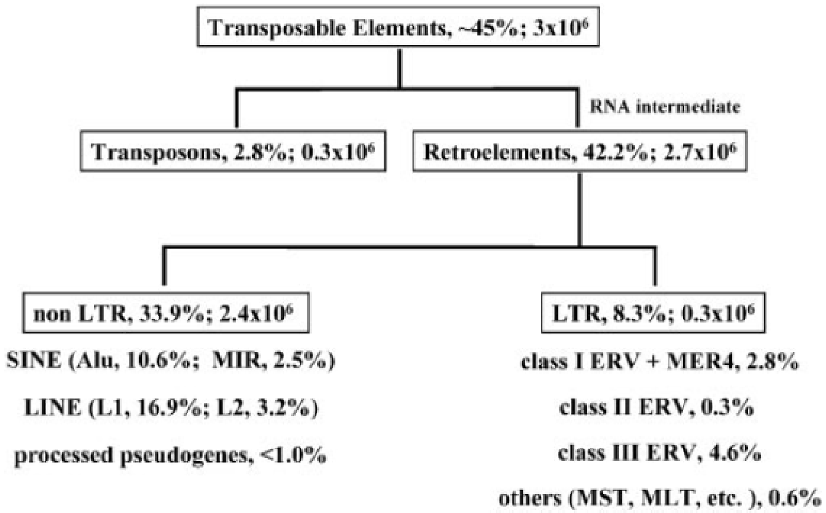
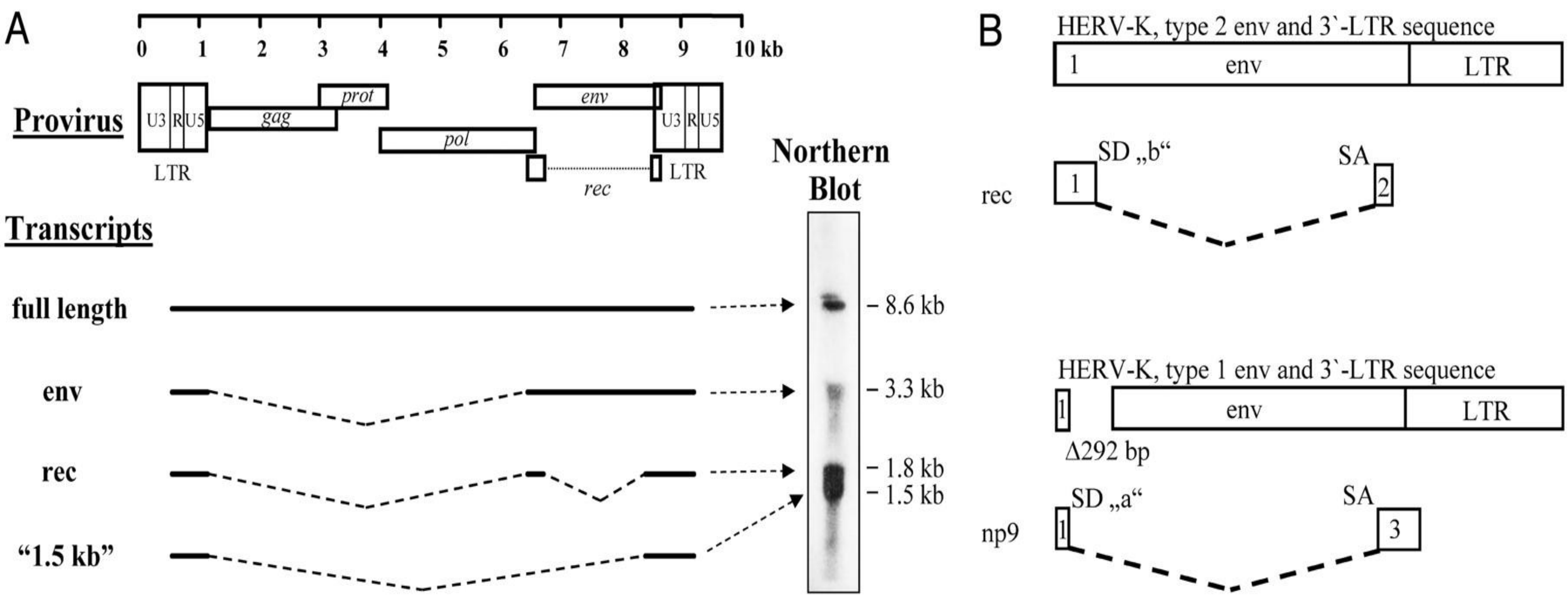


Fig. 1. Classification of transposable elements. The percentage of each element in the genome and the estimated number of the elements of the main groups are indicated.

Bannert, N., & Kurth, R. (2004). Retroelements and the human genome: new perspectives on an old relation. *Proceedings of the National Academy of Sciences*, *101*(suppl_2), 14572-14579.

Fig. 3. Genomic organization and transcript pattern of HERV-K

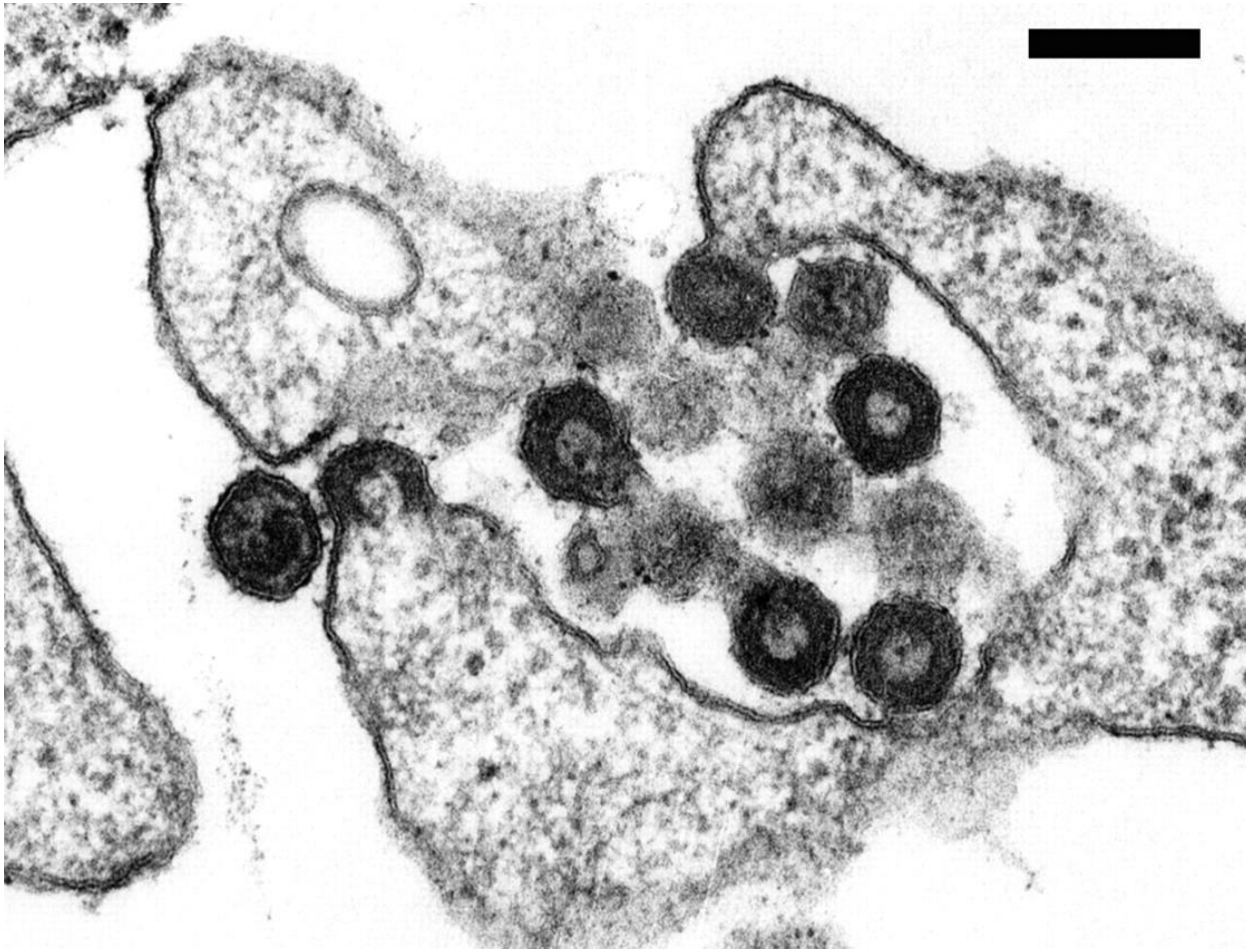


Bannert, Norbert and Kurth, Reinhard (2004) Proc. Natl. Acad. Sci. USA 101, 14572-14579

a. Genome organization of HERV-K

PNAS

Fig. 4. Electron microscopic analysis of HERV-K/HTDV particles produced by teratocarcinoma cell lines



Bannert, Norbert and Kurth, Reinhard (2004) Proc. Natl. Acad. Sci. USA 101, 14572-14579

Cell

Leading Edge
Review

Human Transposon Tectonics

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Mobile DNAs have had a central role in shaping our genome. More than half of our DNA is comprised of interspersed repeats resulting from replicative copy and paste events of retrotransposons. Although most are fixed, incapable of templating new copies, there are important exceptions to retrotransposon quiescence. De novo insertions cause genetic diseases and cancers, though reliably detecting these occurrences has been difficult. New technologies aimed at uncovering polymorphic insertions reveal that mobile DNAs provide a substantial and dynamic source of structural variation. Key questions going forward include how and how much new transposition events affect human health and disease.

majority, about 214,000, corresponded specifically to transposable elements. Their data also suggest a high degree of spatio-temporal specificity and correlation between transposon-initiated transcription and expression of proximal genes. This suggests that coregulation of repeats and neighboring loci supersedes transcriptional interference.

Neoplastic conditions will rightly receive special attention as functional impacts of the human mobilome are approached

Concluding Remarks

High copy number, self-propagating repeats are landscape-determining components of our DNA. Many are fixed, static, and slowly eroding features, but there are also focal "hot spots"—elements that demonstrate episodic activity and sites where new insertions can be found. Today, new technologies are being developed to reveal insertions, confirming that interspersed repeat polymorphisms are important sources of genetic

variation in human populations and suggesting that each of us has somatic compartments genetically variegated by insertion events. Many models of how repeats may influence human phenotypes are emerging, and we look forward to delineating and altering functions of specific insertions in human disease.

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